

KING FAHD UNIVERSITY OF PETROLEUM & MINERALS

Electrical Engineering Department

Senior Design Project



**EE 411
Term # 172
Section #13**

Prototype of a Wireless Seismic Data Acquisition System

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Abstract

Wires are used to conduct seismic surveys to differentiate between the subsurface earth layers and monitor their composition. High cost and heavy loads are introduced by such wires when the number of sensors becomes very large. To avoid these drawbacks, seismic surveys are in research progress to convert the system to be wireless. Our final design receives the seismic signal from coil-based geophone and pass it through conditioning circuits to enhance it before converting it to digital words. The obtained bits are then sent wirelessly to a receiver which stores and process them. The designed system (conditioning circuit and wireless communication) was tested and verified with readably reliable results. The results gave an insight of possible future design upgrading.

Acknowledgements

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List of Abbreviations

AAF:	Anti-Aliasing Filtering
ADC:	Analogue to Digital Converter
API:	Application Programming interface
ASIC:	Application-Specific Integrated Circuit
BPF:	Band Pass Filter
FFT:	Fast Fourier Transform
FPGA:	Field Programmable Gate Array
GC:	Gain Control
GCX:	Geospace Combined Recorder
GFSK:	Gaussian Frequency Shift Keying
HDL:	Hardware Description Language
HPF:	High Pass Filter
I/O:	Input / Output
I2C:	Inter-Integrated Circuit
IDE:	Integrated Development Environment
IEEE:	Institute of Electrical and Electronics Engineering
ISM:	Industrial, Scientific and Medical
KFUPM:	King Fahd University of Petroleum and Minerals
LPF:	Low Pass Filter
LP-WPANs:	Low-Rate Wireless Personal Area Networks
MEMS:	Micro-Electro-Mechanical Systems
MNoGs:	Maximum Number of Geophones
PAN:	Personal Area Network
S.C.C. :	Signal Conditioning Circuit
SAR:	Successive Approximation Register
SNR:	Signal to Noise Ratio
SPI:	Serial Peripheral Interface
UART:	Universal Asynchronous Receiver-Transmitter
USART:	Universal Synchronous and Asynchronous Receiver-Transmitter

Chapter 1

Introduction

This project is intended for senior standing undergraduate students who are taking the course EE411 (Senior Design Project) as a group at King Fahd University of Petroleum & Minerals (KFUPM). To implement the gained conceptual ideas and skills from the attended lectures and laboratories, a wireless prototype of a seismic data acquisition prototype is implemented. Such recently developed practical application combines many aspects of electrical engineering such as electronics, communication, microcontrollers, signal processing and control engineering. A brief overview will be given (Section 1.1) to introduce the concept and the essential issues related with the commonly used seismic data acquisition systems. Problem definition (Section 1.2) and objectives (Section 1.3) will be stated as well. Finally, the project management details (Section 1.4) will be listed.

1.1 Overview

Acquiring geophysical data (usually underground mapping) consist of the following stages: Sensing, Recording, Transmission and Receiving. Data sensing is mainly done using the concept of induced current in a damped oscillator. However, a new technology based on microelectromechanical systems (MEMS) is starting to emerge as an alternative solution. The recording stage determines the degree of accuracy and should have a lower and an upper limit on how much memory bits are allocated for digitalization without any loss of useful data. Transmission and receiving, can be done in two ways: wired or wireless. Until recently, cables were used to propagate the data from the sensing device to a central unit (storage and Processing unit). Wireless communication was suggested to compensate the inefficiency of wired systems especially when the number of data sources became extremely large. Seismic data acquisition is a reliable method for identifying the underground subsurface layers and monitoring oil and gas resources. The process demands a large number of sensors distributed over wide area (typically, more than 40 km²). Such surveys require +300,000 seismic sensors wired together as grouped clusters delivering the acquired data to the central unit. Approximately, cables cost 50% of the exploration expenses

and 75% of the total equipment weight. Our design suggests a solution to this problem by transferring the acquisition process to be wireless. Using components with low power consumption feature which can be integrated in a single board resulting in a compact practical design that is cheaper alternative to the available cabled system. [1]

1.2 Problem Definition

The used seismic data acquisition systems for oil and gas explorations are costly and their deployment requires excessive manpower. These issues are mostly related to the used cables throughout large seismic surveys. Wireless communication faces many challenges such as achieving acceptable throughput, power consumption and relative cost.

1.3 Objectives

1. Build a wireless seismic data acquisition prototype with emphasizes on low power consumption.
2. Examine different possible seismic sensor.
3. Implement the required conditioning circuits.
4. Test different wireless communication techniques.

1.4 Project Management

This section shows the team organization process as the time line for the whole project. Table 1 shows the team management cycle throughout the semester. Microsoft project management was used in order to avoid any late tasks and submissions. Samples of some of the used schedules and tables are shown in Figure 1 and Figure 2.

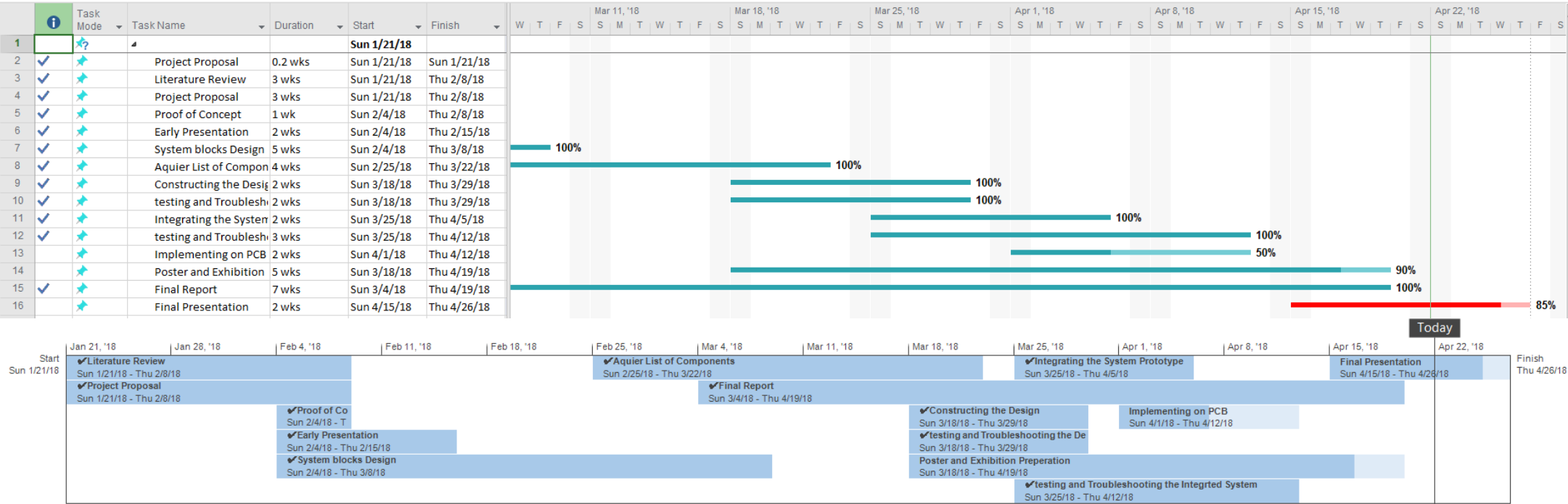
Table 1. Team management.

No.	Student Name	Student ID	Main Tasks	Leadership Period (Weak #)
1.	Abdullah Al-Nafisah	201364070	➤ Sensors ➤ ADC	13 - 15
2.	Faisal Al-Masri	201228500	➤ Project Management ➤ ADC	-
3.	Feesal Swaid	201248380	➤ Microcontrollers	-
4.	Mohanad Al-Mousa	201325670	➤ Pre-ADC conditioning	9 - 12

No.	Student Name	Student ID	Main Tasks	Leadership Period (Weak #)
5.	Mojtaba Al-Shams	201337890	➤ Wireless Communication	4 - 8

Task Name	Duration	Start	Finish	% Complete
Senior Project	13.8 wks	Sun 1/21/18	Thu 4/26/18	100%
Literature Review	3 wks	Sun 1/21/18	Thu 2/8/18	100%
Project Proposal	3 wks	Sun 1/21/18	Thu 2/8/18	100%
Proof of Concept	1 wk	Sun 2/4/18	Thu 2/8/18	100%
Early Presentation	2 wks	Sun 2/4/18	Thu 2/15/18	100%
System blocks Design	5 wks	Sun 2/4/18	Thu 3/8/18	100%
Aquier List of Components	4 wks	Sun 2/25/18	Thu 3/22/18	100%
Constructing the Design	2 wks	Sun 3/18/18	Thu 3/29/18	100%
testing and Troubleshooting the Design	2 wks	Sun 3/18/18	Thu 3/29/18	100%
Integrating the System Prototype	2 wks	Sun 3/25/18	Thu 4/5/18	100%
testing and Troubleshooting the Integrted System	3 wks	Sun 3/25/18	Thu 4/12/18	100%
Implementing on PCB	2 wks	Sun 4/1/18	Thu 4/12/18	100%
Final Presentation	2 wks	Sun 4/15/18	Thu 4/26/18	100%
Poster and Exhibition Preperation	5 wks	Sun 3/18/18	Thu 4/19/18	100%
Final Report	7 wks	Sun 3/4/18	Thu 4/19/18	100%

Figure 1. Microsoft project management.



PROJECT OVERVIEW

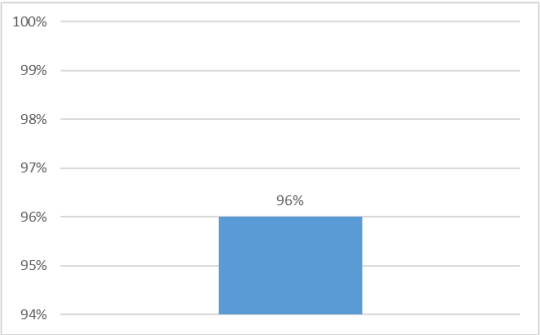
SUN 1/21/18 - THU 4/26/18



MILESTONES DUE
Milestones that are coming soon.

Name	Finish
------	--------

% COMPLETE
Status for all top-level tasks. To see the status for subtasks, click on the chart and update the outline level in the Field List.



LATE TASKS
Tasks that are past due.

Name	Start	Finish	Duration	% Complete	Resource Names
Implementing on PCB	Sun 4/1/18	Thu 4/12/18	2 wks	50%	
Poster and Exhibition Preparation	Sun 3/18/18	Thu 4/19/18	5 wks	90%	

Figure 2. Microsoft project management.

Chapter 2

Background

Not long ago, wireless seismic data sensors have been developed in the hope of satisfying the customers' needs. Many products are found in the market falling under application dependent categories. For example, the seismic sensing industry includes: oil and gas exploration, seismic hazards anticipation, crowd identification and stability tests. The focus of this chapter is to explore the existing products that are intentional for the petroleum industry. More specifically, wireless seismic data acquisition systems will be identified. Section (2.1) presents commercially available related products. Their features are then analyzed and compared in section (2.2). Section (2.3) concludes this chapter with detailed technical data that covers the essential component of the desired system.

2.1 Existing Product

In the petroleum industry, downstream companies commercialize fully developed wireless seismic sensing systems. These devices consist of channels for external sensors (or a single internal sensor) along with the needed circuitry for conditioning and propagating the acquired data to the next unit. The list of the found companies is very long from which three different companies were selected based on the available data to investigate their products in this section: Sercel, FairfieldNodal and GEOSPACE Technologies. Sercel is well known in the petroleum industry and provides all kinds of seismic monitoring solutions in general. They developed multi-environments seismic sensing node called UNITE [2] with two versions: eX (accepts 1 sensor) and eX-3 (accepts three sensors). No information was given for the number of bits corresponding to the used ADC chip in their product. The product from FairfieldNodal company ZLAND 1C [3] integrates all of the required circuitry for a single internal sensor (option for external sensor is available as well) except for the wireless communication which is assumed to be provided by the customer using external connectors. The autonomous recording cable-free Geospace Combined Recorder (GCX) [4] node is being commercialized by GEOSPACE Technologies. Incorporating a single bulky device, GCX was made to store the acquired data for long period of time in a solid-state flash memory without the need of wireless transmission. However, it is not practical in case

of oil and gas exploration because each GCX can process only a single channel. Table 2 presents the claimed specifications of the given features within each product from the three companies.

Table 2. Comparison of existing system wise products

Feature	Claimed Specification		
	UNITE (Sercel)	ZLAND 1C (FairfieldNodal)	GCX (GEOSPACE Technologies)
Sensor	-	Natural frequency = 10 Hz Damping = 0.7 Sensitivity = 78.7 V/(m/s)	Sensitivity = 78.7 V/(m/s)
ADC Resolution	-	24 bits	24 bits
Sampling Interval	2 ms	0.5, 1, 2, 4 ms	0.25, 0.5, 1, 2, 4 ms
Data rate	Up to 11 Mbps	-	-
Wireless Carrier Frequency	2.4-2.4835 GHz	-	-
Range	Up to 1500 m (Omni-directional antenna)	-	-
Preamplifier Gain	-	0-36 dB (step = 6 dB)	0-36 dB (step = 6 dB)
Anti-Alias Filter	-	206.5 Hz (at 2 ms sampling) (82.6% of Nyquist)	83% of Nyquist
DC Blocking Filter	-	1-60 Hz (step = 1 Hz)	-

Feature	Claimed Specification		
	UNITE (Sercel)	ZLAND 1C (FairfieldNodal)	GCX (GEOSPACE Technologies)
Frequency Response	-		1-1650 Hz (0.25 ms sampling)
Battery	Rechargeable 130 h (continuous operation)	Rechargeable Li- Ion 960 h (continuous operation) (at 2 ms sampling)	720 h (continuous operation)
Weight	1.6 kg (eX), 1.95 kg (eX-3)	1.8 Kg	2.72 Kg
Maximum Input Signal	-	-	1.8 Vrms
Equivalent Input Noise	-	-	0.17 μ Vrms
Instantaneous Dynamic Range	-	-	124 dB (2 ms sampling)
GPS	Clock synchronization (better than 20 μ s)	-	-

2.2 Technical Data

2.2.1 Sensors

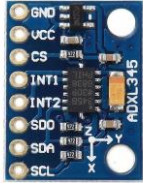
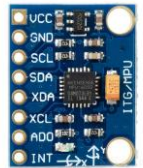
Seismic data for oil and gas explorations are usually collected using what is called Coil-Based Geophones. Micro-Electro-Mechanical Systems (MEMS) recently showed the high potential of entering such industry due to the apparent technological advancement [5]. In case of MEMS, sensing mechanical vibrations by translating acceleration into electric signals is usually done by one of three possibilities: (1) Capacitive, (2) Piezo-

electric and (3) Piezo-resistive. When selecting a seismic sensor, the following parameters are considered [5]:

1. Output type: analog or digital.
2. Number of axis: 1 or 3.
3. Frequency response.
4. Power consumption.
5. Measurement range: Input amplitudes with ensured linearity.
6. Sensitivity: output produced from the applied Input.
7. Dynamic range: maximum to minimum measurable signals (similar to SNR).
8. Shock limit: Maximum input after which the device is damaged.

It is desired to have a flat frequency response within the acquired signal frequency contents. Sensitivity is a measure of how much output signals can be obtained for a given mechanical input (voltage per speed unit in the case of Coil-Based Geophones). The dynamic range of sensors in general is defined as the largest divided by the smallest resolvable input signals resulting (in most cases) in large quantities which are more appropriately represented by logarithmic scales in dB units. Several sensors were selected for further investigation are shown in Table 2.

Table 3. Comparison between different vibration sensors.

Sensor	Picture	Output type	Number of axes	Sensitivity (resolution)	Power Consumption	Measurement range	Shock limit	Band width	Damping constant	Structure
SM-24		Analog	1	28.8 V/(m/s)	Not mentioned	Not mentioned	Warranty period is 3 years.	10 to 240 Hz	0.25 (open circuit)	Fixed magnet and a moving case rounded with a coil. The moving case weight is 11 grams.
GY-291 ADXL345		Digital (up to 13-bit)	3	4 mg/LSB	Measurement mode (23uA) and Standby mode (0.1uA)	Up to $\pm 16g$	10,000g	DC to 0.1 or 3200 Hz (user selectable)	Not mentioned	Deflection of the structure is measured using differential capacitors that consist of independent fixed plates and plates attached to the moving mass.
GY-521 MPU-6050		Digital (16-bit)	3	Accelerometer 2 to 16 LSB/g Gyroscope 16 to 131 LSB/(°/s)	Depends on the operating conditions.	Accelerometer ($\pm 2g$, $\pm 4g$, $\pm 8g$, and $\pm 16g$). Gyroscope (± 250 , ± 500 , ± 1000 , and ± 2000 degree/sec)	10,000g	Not mentioned	Not mentioned	Separate proof masses for each axis and capacitive sensors detect the displacement differentially.

Sensor	Picture	Output type	Number of axes	Sensitivity (resolution)	Power Consumption	Measurement range	Shock limit	Band width	Damping constant	Structure
LDT0-028K		Analog	Typically, 1 but multi-axis response can be achieved by positioning the mass off center.	Baseline: 50 mV/g At resonance: 1.4 V/g (both increased by adding small masses to the tip of the sensor)	Not mentioned	Not mentioned	Not mentioned	Resonant frequency: 180 Hz (decreased by adding small masses to the tip of the sensor) Flat response up to 90 Hz	Not mentioned	28 μ m thick piezoelectric PVDF polymer film with screen-printed Ag-ink electrodes, laminated to a 0.125 mm polyester substrate, and fitted with two crimped contacts.
MiniSense 100		Analog	Not mentioned	Baseline: 1 V/g At resonance: 5 V/g	Not mentioned	Not mentioned	Withstands High Shock	Resonant frequency: 75 Hz Flat response up to 40 Hz	Not mentioned	Cantilever-type vibration sensor loaded by a mass. The active sensor area is shielded for improved RFI/EMI rejection. PVDF sensing element.

The frequency response of SM-24 (Coil-Based Geophone) can be seen in Figure 3 [6].

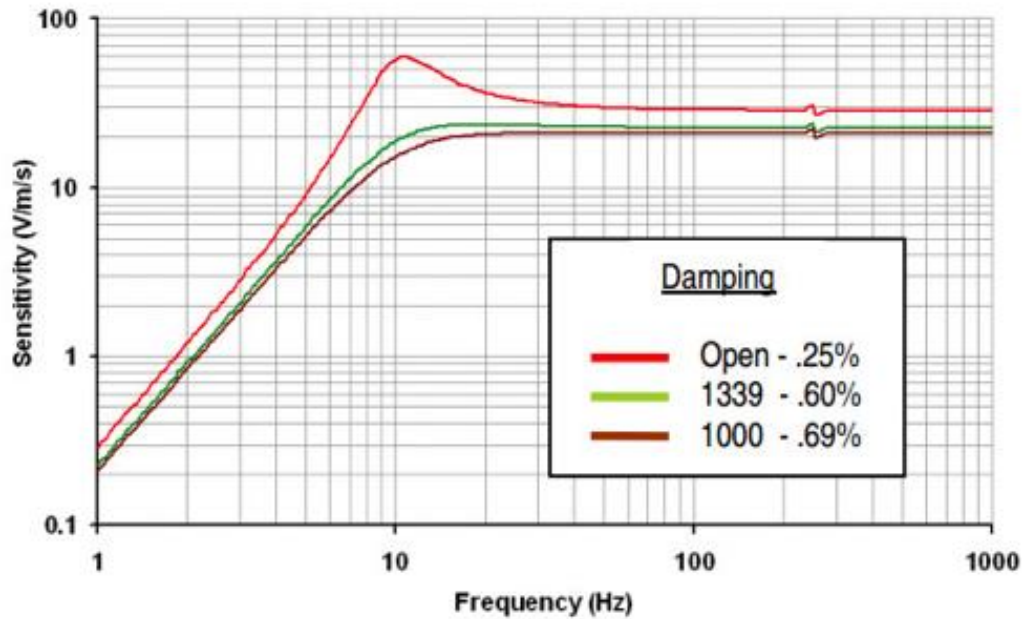


Figure 3. SM-24 Frequency response.

The block diagram of the GY-291 ADXL345 module is shown in Figure 4 [7].

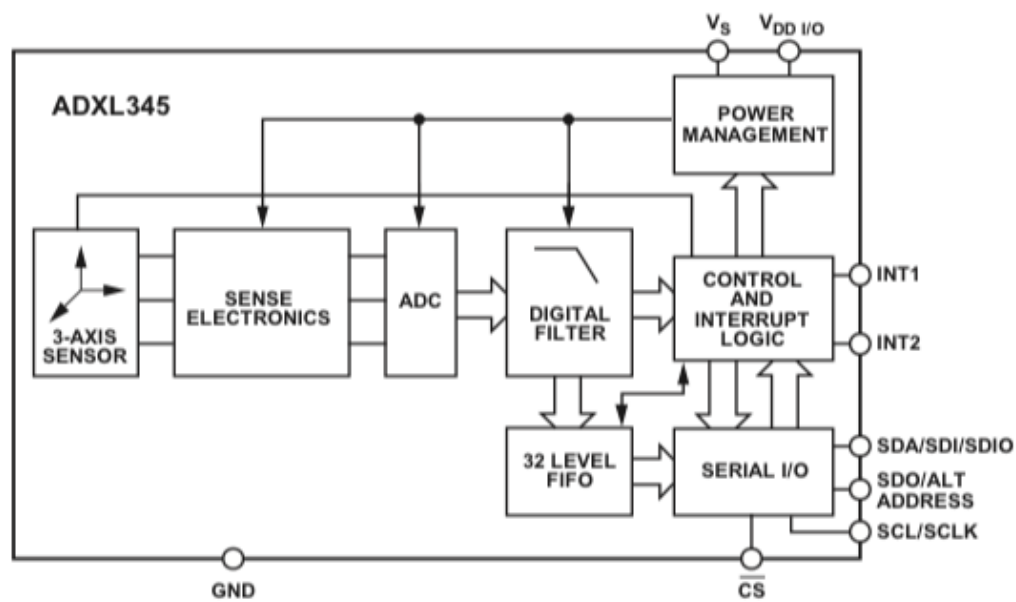


Figure 4. GY-291 ADXL345 block diagram.

The GY-521 MPU-6050 sensor module integrates two MEMS devices in a single chip: an accelerometer and a gyroscope. MEMS gyroscope senses any rotation with respect to each axis. By utilizing the Coriolis effect, capacitive pickoff can measure the observed vibration. The resulting signal (proportional to angular rate) is amplified,

demodulated, and filtered before going to the ADC stage. The accelerometer employs three proof masses for each axis separately (with separate ADCs). Capacitive sensing is used to detect the differential displacement of the proof masses. Under normal conditions (when the chip is vertically placed), it will measure 0g on the x and y axes and +1g on the z axis. The block diagram is presented in Figure 5. [8]

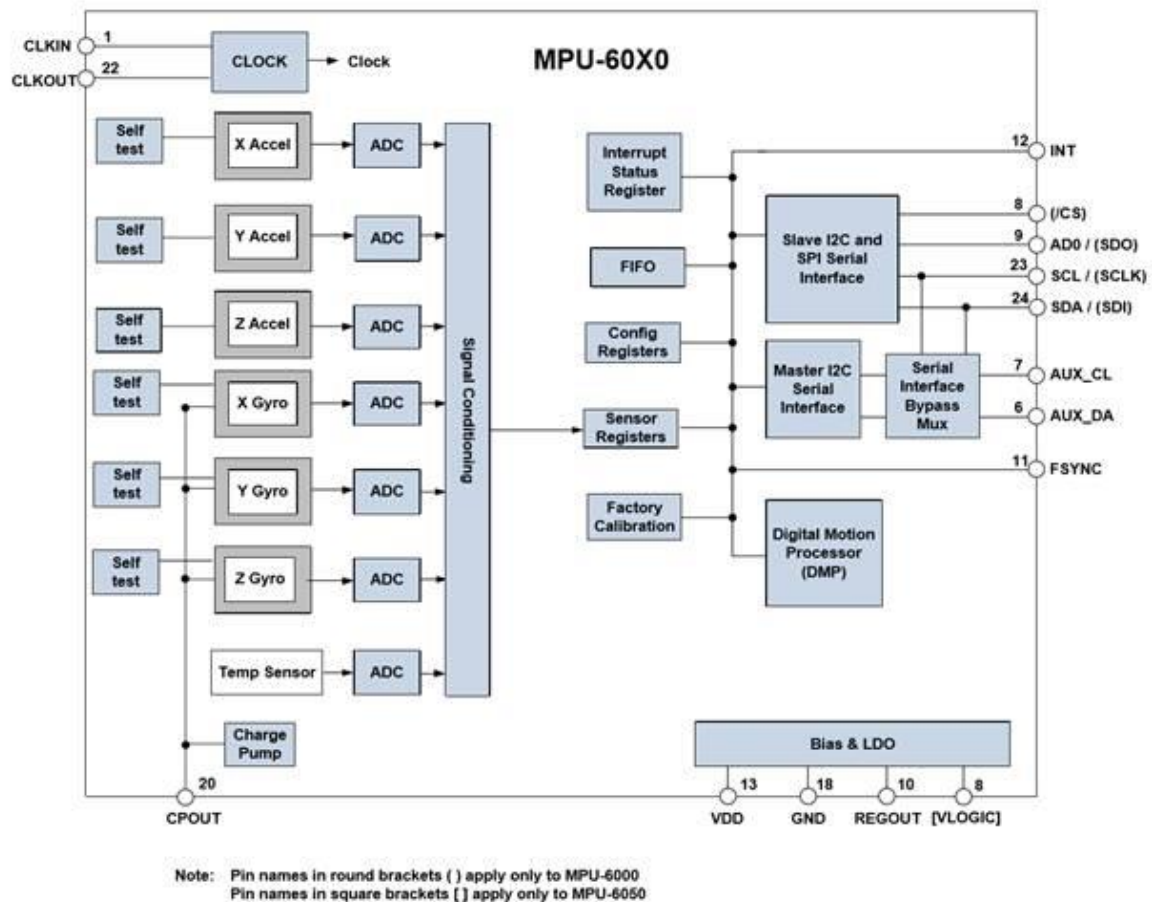


Figure 5. GY-521 MPU-6050 block diagram.

The Piezoelectric vibration sensor (LDT0-028K [9]) normalized dimensions can be seen in Figure 6. Figure 7 shows its frequency response where masses are attached to the sensor. The heavier the attached mass the higher the cutoff frequency of the frequency response with less sensitivity (indicated by the bolder lines in the figure).

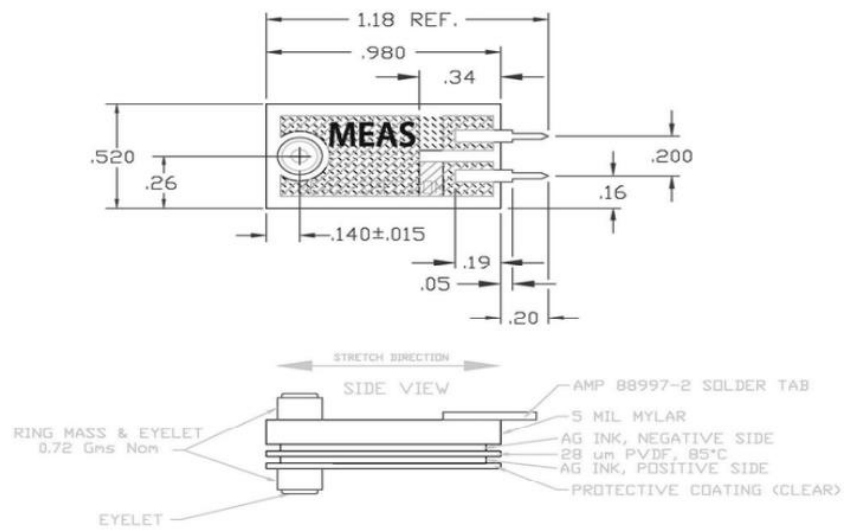


Figure 6. LDT0-028K dimensions.

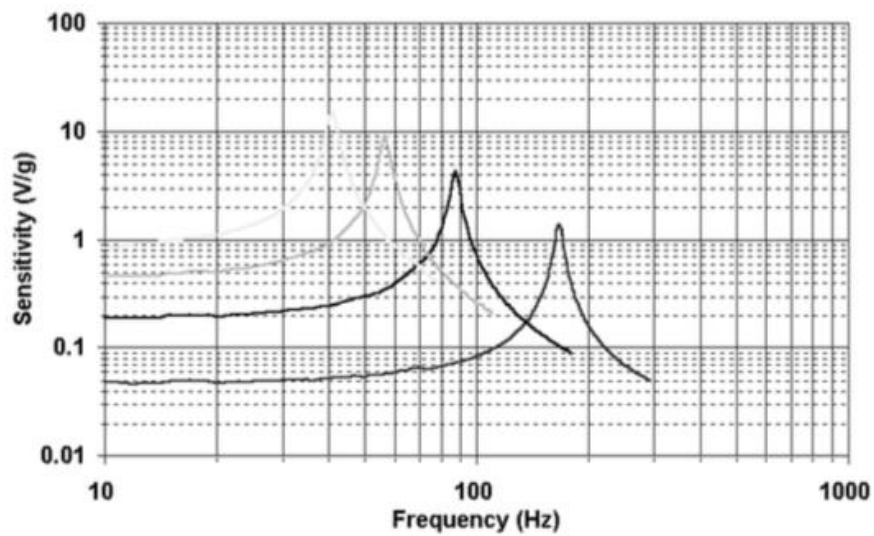


Figure 7. LDT0-028K frequency response.

2.2.2 Conditioning Circuits

Before converting the analog signal into digital, appropriate conditioning is required. Amplifiers in general provide the gain and can be categories as:

1. Active devices.
2. Power amplifiers.
3. Operational amplifiers (op-amps).
4. Distributed amplifiers.
5. Switched mode amplifiers.

The possible stages of such amplifiers are classified into common terminal, unilateral or bilateral and inverting or non-inverting. Amplifying the signals without considering SNR is not beneficial. Therefore, matching filter is required to limit the noise by decreasing the BW of the channel and increase the SNR. The continuous nature of our world forced scientists and engineers to come up with innovative ways of transforming analog electrical signals into discrete and digital words of ones and zeros. As a result, analog to digital converting stage (ADC) is essential. With an anti-aliasing filtering(AAF), the amount of amplification aims to match the SNR of the analog input to the ADC levels. The concept of ADC can hold a degree by its own due to the wide variety of configurations and technologies such as Flash, Successive Approximation Register (SAR), Dual Slope and Delta-Sigma. these four types will be investigated shortly to understand the arguments behind chosen the Delta-Sigma ADCs for seismic monitoring applications in general from a practical point of view. Flash ADC is the oldest (usually known as a direct-conversion) uses a linear voltage ladder with a comparator at each ladder to compare the input voltage to successive reference voltages. Flash ADCs are made by cascading high speed series connected comparators. Each comparator compares V_{in} to a different reference Voltage as shown in Figure 8. For an N-bit converter, the circuit needs 2^{N-1} comparators and 2^N resistors to provide the reference voltage.

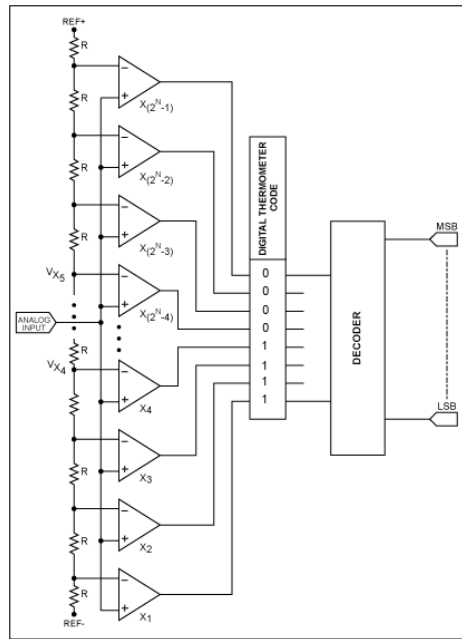


Figure 8. Flash ADC illustrative diagram.

In case of the SAR ADC, Analog waves are converted into a sequence of bits by searching through all possible quantization levels before eventually converging upon a serial digital output for each conversion cycle. As can be seen from Figure 9, a single comparator is used to compare the most significant bit of the device in its analog form (generated from the 1-bits DAC) with the input voltage. If the input signal is larger, a 1 is produced and saved within the SAR register. The cycle is repeated to identify the remaining bits until it reaches the least significant where a complete word is extracted serially.

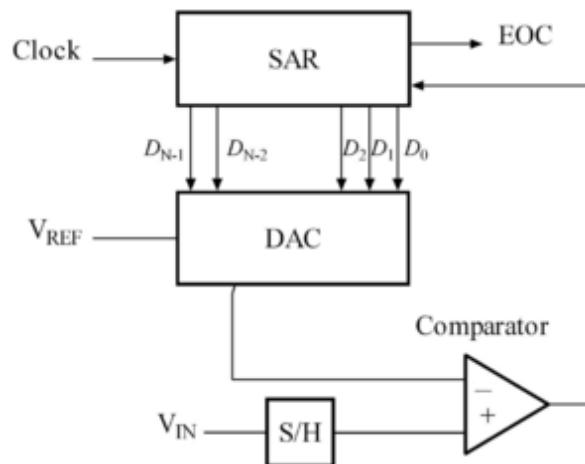


Figure 9. SAR illustrative diagram.

Dual Slope ADC type utilizes the slope of two signals: the input voltage and a negative reference voltage. Figure 10 demonstrates the circuit of a dual slope ADC. Moving the switch to the input signal (V_A) will charge the capacitor in an approximately linear slope. Turning the switch to the negative reference (V_{ref}) discharges the capacitor which allows a precise digital conversion of the input voltage.

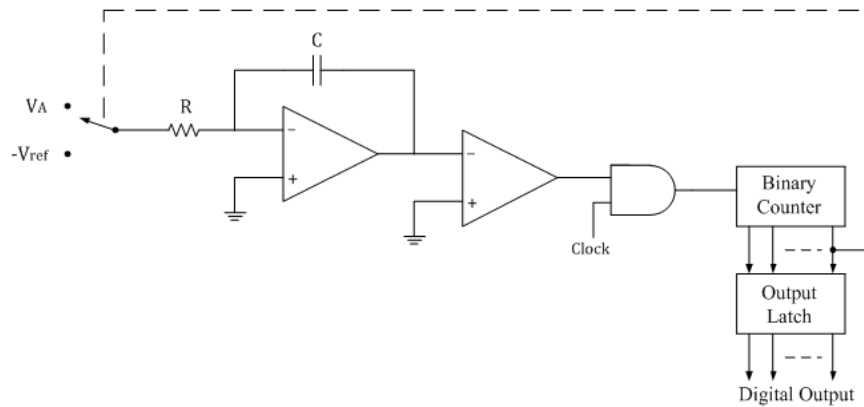


Figure 10. Dual Slope ADC illustrative diagram.

Finally, Δ - Σ ADC accumulates large number of samples in time (oversampling) and then takes the average of them to produce a single word of bits. Hence, input signals need to be relatively slow in order to obtain very accurate conversions. An illustrative diagram is shown in Figure 11.

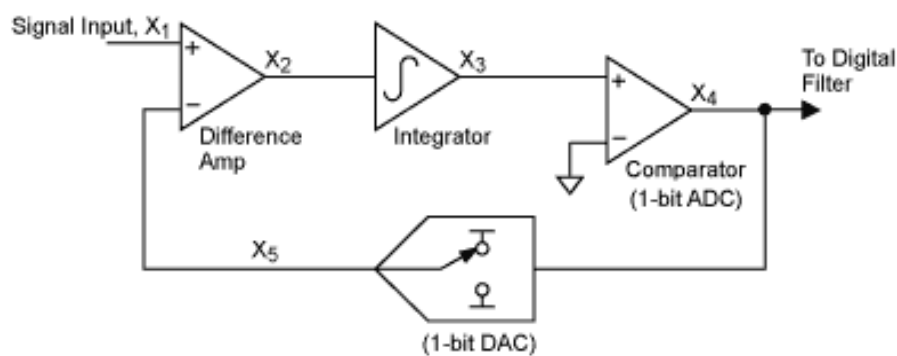


Figure 11. Δ - Σ ADC illustrative diagram.

Table 4 compares the apparent advantages and disadvantages of the considered ADC types. Now, it is clear why Δ - Σ ADC is utilized in seismic data acquisition. Its high resolution for low frequencies (where seismic data are concentrated) along with their

low power consumption and low cost make it the optimum available ADC type (because of the large number of the used sensors).

Table 4. Advantages and disadvantages of ADC types.

Type	Advantages	Disadvantages
Flash	➤ Fast (almost instantaneous). ➤ Simplicity.	➤ Requires huge number of parts (exponentially proportional to the number of bits). ➤ Low resolution (mainly caused by the mismatch between parts) ➤ High power consumption
SAR	➤ Constant power dissipation versus sampling rate. ➤ High resolution.	➤ Slow conversion. ➤ Easily subjected to noise causing inaccurate readings.
Dual Slope	➤ High accuracy. ➤ Immune to noise (coined to the conversion differential nature).	➤ Slow conversion ➤ Expensive. ➤ Accuracy may be affected by improper external connections.
Δ - Σ	➤ High resolution. ➤ Low power consumption. ➤ Low cost.	➤ Not practical for high frequency applications due to oversampling.

2.2.3 Signal Processing

Signals are now surrounding us everywhere. Those signals carry intensive information about the world nature and life. To understand those signals, there must be ways to store and manipulate them. Microcontrollers are devices that have the capabilities to store data, and with some logical connections and algorithms, they can manipulate, and store signals obtained using sensors. In addition, Field Programmable Gate Array (FPGA) is another type of devices that is utilized to manipulate signals. ATtiny85, AT89s8253, and Atmel328P are such microcontrollers that can be used for this purpose in addition to Nexys3 which uses FPGA.

ATtiny85

ATtiny85 is an 8-bit microcontroller based on the AVR enhanced RISC architecture provided by Atmel. It has low power consumption and small size making it suitable to reduce project sizes. It has 8 pins and 5 of which can be used for programming and it can be reprogrammed, and a memory of 8KB and 6 I/O pins. Figure 12 shows the physical structure of the chip.



Figure 12. ATtiny.

AT89s8253

AT89s8253 is an 8-bit microcontroller provided by Atmel with high performance and low-power consumption. It has 12K bytes memory and 4 I/O ports each of 8 pins. It has SPI and UART interfaces.

ATmega328P

ATmega328P is an 8-bit microcontroller provided by Atmel with low power consumption. It has 3 ports each with 8 pins except one. It has Serial Peripheral Interface (SPI), universal synchronous and asynchronous receiver-transmitter (USART), Inter-Integrated Circuit (I2C) interfaces.

Nexys3

The Nexys3 is a platform for digital circuit development that is based on the Xilinx Spartan 6 LX16 FPGA. The Spartan 6 is optimized to have high performance logic. FPGA stands for a field programmable gate array is a programmable integrated circuit. It can be configured using a hardware description language (HDL), similar to the one used for an application-specific integrated circuit (ASIC). The physical shape is shown in Figure 13.

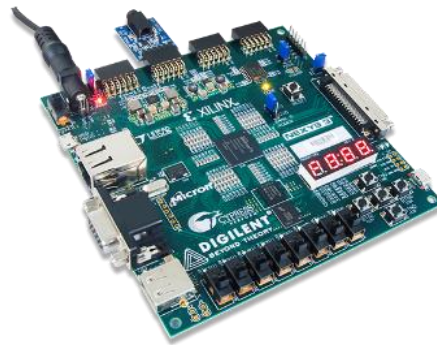


Figure 13. Nexys3.

2.2.4 Wireless Communication Modules

nRF24L01+

Shown in Figure 14, nRF24L01+ is a transceiver chip provided by Nordic Semiconductor that can operate as transmitter and receiver. It operates at 2.4 GHz within the worldwide industrial, scientific and medical (ISM) bands. It has data rates of 250k, 1M and 2M bps and 126 channels and can use 6 of them together. It uses Gaussian Frequency Shift Keying (GFSK) modulation and has low power consumption.



Figure 14. nRF24L01+ transceiver.

Digi XBee RF Module

Digi XBee S2C RF Module is a wireless communication module provided by Digi International that is operating within the ISM at 2.4 - 2.5 GHz and has low power consumption with line-of-sight range of about 1.2 km. It has maximum data rate of 250 kbps and 5Mbps when using, respectively, UART and SPI serial communication interfaces. It also supports Point-to-point, point-to-multipoint, peer-to-peer and DigiMesh network topologies. The physical shape of the module can be seen in Figure 15.



Figure 15. Digi XBee RF Module.

Chapter 3

Design and Development

This chapter shows the design structure (Section 3.1) and the analysis (Section 3.3) that were needed for the design of project. It also presents the applied constraints and standards (Section 3.2).

3.1 Concept Design

The formal building blocks of any wireless data acquisition system (including seismic data) can be arranged as shown in Figure 16. The Sensor stage determines the system degree of accuracy and should be handled with caution in order to obtain the data with high fidelity. Properly interfacing the sensor's analog signals with the system and suppressing all types of introduced noise are the main functions of the Conditioning Circuits block. Digital data are then transmitted through air within the Wireless Communication block. The Signal Processing unit is present to manage the system operations and regulates the power beside other possible extra features such as initial signal processing. The following sections will explain each block in further details.

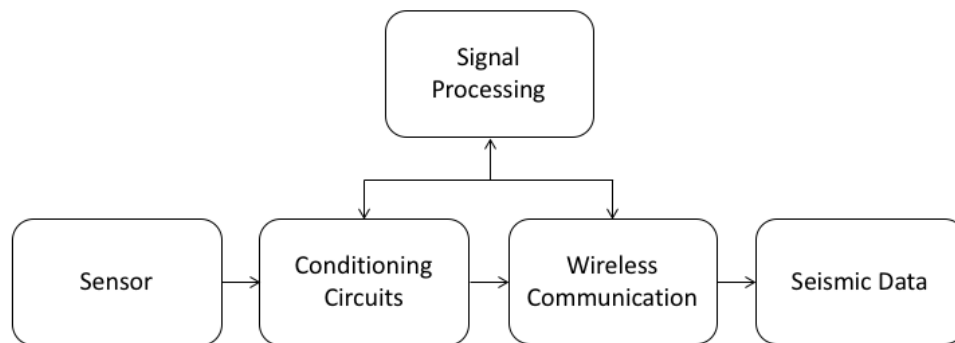


Figure 16. System building blocks.

3.1.1 Sensor

Vibration sensors are categorized into passive and active. Seismic data for petroleum explorations commonly acquired using the concept of induced current in passive damped oscillators known as Coil Geophones. Generally speaking, a Coil Geophone is a fixed magnet surrounded by a cylindrical moving case that is rounded by a coil. From

Faraday's law of induction, electrical signals are generated in direct proportionality to motion of the coil rounded case with respect to the fixed magnet. The natural frequency of a damped oscillator identifies the lowest measurable analog signals. In the case of passive sensors, the upper limit is 0.03Hz [5]. Recently, active sensing using Accelerometers emerged to the field of seismic data as an alternative solution, but it is still under extensive research. Comparison between Coil Geophones and Accelerometers was done theoretically and experimentally in [10], [11] to obtain approximate models for simulation purposes.

3.1.2 Conditioning Circuits

In order to identify and analyze the seismic signals properly, active filtering circuit is used to limit the channel BW the signal from low and high noise signals with critical signal to noise ratio (SNR). To recognize the main signal further, low noise amplifier with high precision is required with different gains according to the strength of the seismic signal. Signal conditioning circuit, is the combination of matching and antialiasing filters and amplification circuits as well as the clipper circuit and the Analog to Digital Converter (ADC). Its importance relies on separating unnecessary signals, mainly associated noise signals, to purify the signal before going to the ADC. ADC transforms the analog data into digitally process-able words of ones and zeros. Converting the analog seismic signal coming after the preamplifier into digital words is done using Delta-Sigma (24-bits) ADCs due to their high accuracy and the inherited oversampling feature [10], [12] which was verified above in section 2.3.2.

3.1.3 Signal Processing

To process, manipulate data and control a system, a system may need storage and logical units that use algorithms. Such units that can serve the purpose are microcontrollers and FPGA units. Microcontrollers and FPGAs have different capabilities, shapes, number of pins and sizes. Depending on the needs, those parameters can be determined. In a seismic data acquisition system, a node collects signals using geophones. Those data are then stored, probably processed and then sent to a data center for further analysis. Microcontrollers and FPGAs can play a vital role in this part.

3.1.4 Wireless Communication

In wireless seismic data acquisition, an energy harvesting performance of low power RF module is needed. Also, the chosen RF module should have a line-of-sight transition capability of 100m or more because the distance between network wireless nodes is around 100m [1]. Each leaf node (geophone) sends data to the cluster head then data is sent to gateway node to send them to a storage and processing unit via long range (Wi-Fi) communication [1]. The current design is only composed of transmitting and receiving data to process them (similar to cluster head that send data to the storage and processing unit) instead of the complete networking process.

3.2 Realistic constraints and standards

You project must include all realistic constraints and standard related to the project.

3.2.1 Realistic constraints

✓ Technical:

- At least 16 bits ADC conversion.
- Sampling interval of at least 2ms.
- Wireless transmission of at least 50m.
- Only commercially available products can be used (without manufacturing).

✓ Non-technical:

- Total cost of the constructed prototype should not exceed 500 SR.
- Time limitation of only 14 weeks.
- The multidisciplinary nature of the project.

3.2.2 Standards

- IEEE 1241: Analog to Digital Converters [13]
- IEEE C95.1: Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 3 kHz to 300 GHz [14].
- IEEE 754: Standard for Binary Floating-Point Arithmetic [15].
- IEEE 802.15.4: Low-Rate Wireless Personal Area Networks (LR-WPANs) [16].

3.3 Preliminary Analysis

3.3.1 Sensor

Coil-Based Geophone

Two techniques of sensing seismic data have been employed: voltage induction and capacitive sensing. In the voltage induction method, a moving coil surrounding a fixed magnet generates the signals in proportionality to movements. The transfer function from mechanical vibrations to electrical signals was derived in [10], [11] as follows. Starting from Faraday/Lenz law of electromagnetic induction:

$$v_g \propto \frac{dx}{dt} \quad (1)$$

It is assumed that only a sensitivity constant can make the proportionality sign equality (call it A). The coil-based sensor ideally behaves as a simple harmonic oscillator with the following equation:

$$\frac{d^2x}{dt^2} + 2\lambda\omega_o \frac{dx}{dt} + \omega_o^2 x = \frac{d^2u}{dt^2} \quad (2)$$

Where x is the magnet's position with respect to the coil, λ is the damping ratio, ω_o is the natural frequency, u is the coil position relative to the inertial frame and it is time. Transformation to the s domain yields three equations that represent the electrical signals responses to the displacement, velocity and acceleration in the coil-based sensor:

$$H_{displacement}(\omega) = A \frac{s^3}{s^2 + 2\lambda\omega_o s + \omega_o^2} \quad (3)$$

$$H_{velocity}(\omega) = A \frac{s^2}{s^2 + 2\lambda\omega_o s + \omega_o^2} \quad (4)$$

$$H_{acceleration}(\omega) = A \frac{s}{s^2 + 2\lambda\omega_o s + \omega_o^2} \quad (5)$$

Figure 17, Figure 18 and Figure 19 shows the frequency and phase step responses of the above transfer functions using MATLAB (script is attached) with $\omega_o = 10$ (rad/s) and $\lambda = 0.25$ (see Appendix A).

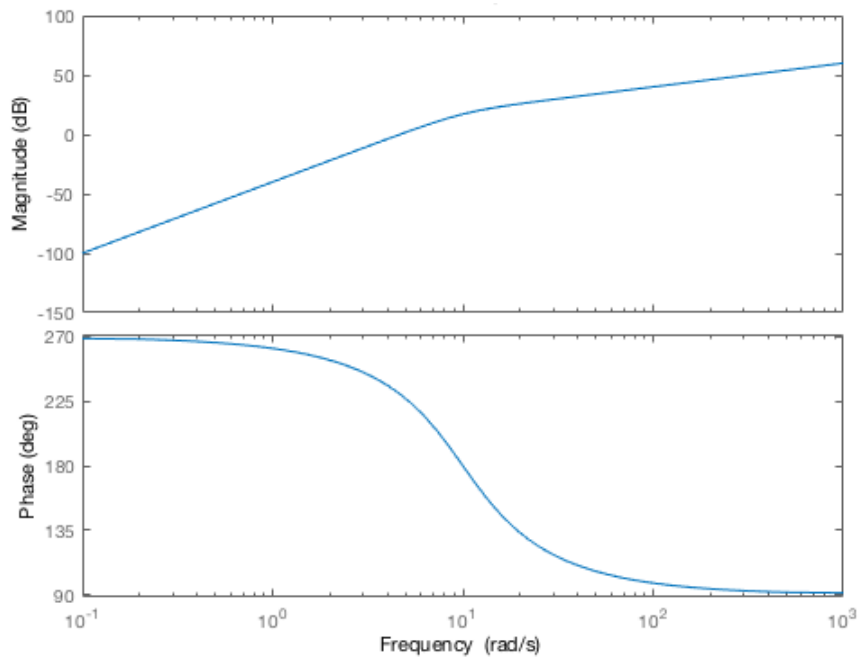


Figure 17. Displacement step response (normalized).

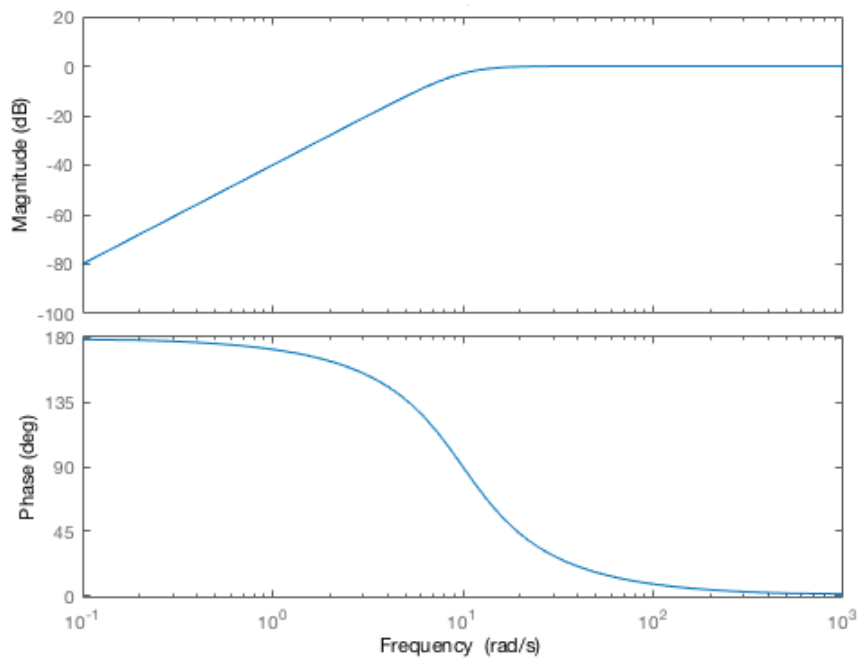


Figure 18. Velocity step response (normalized).

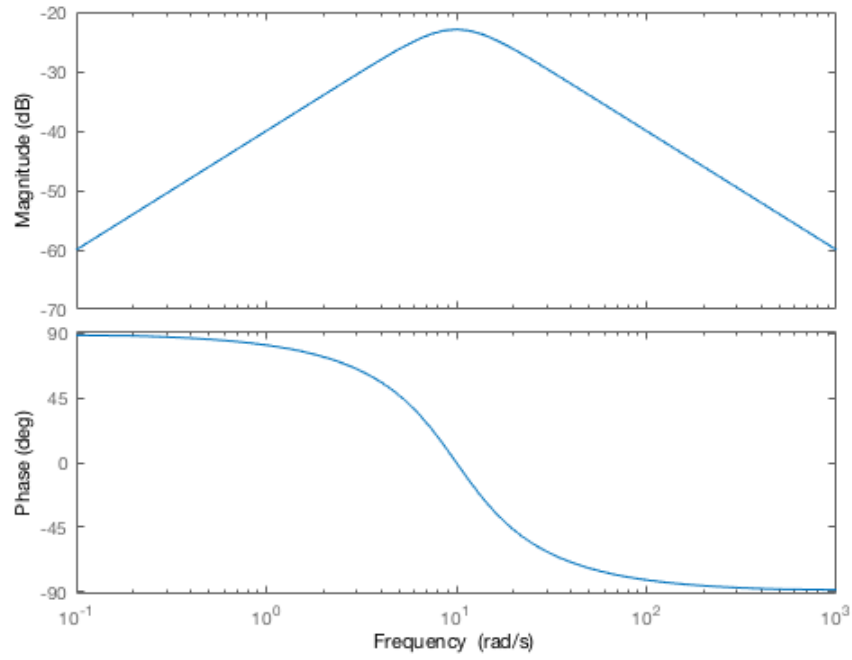


Figure 19. Acceleration step response (normalized).

Due to the approximately flat velocity response of coil-based seismic sensors at frequencies above ω_o , they are commonly known as velocity sensors.

Capacitive Sensors

In [10], derivation of the governing transfer equation of the capacitive type was done starting from the fact that a voltage across a capacitor has linear proportionality to the distance between its terminals:

$$v_m \propto x \quad (6)$$

As in the coil-based sensor, it was assumed that only sensitivity constant can make the proportionality sign equality (call it B) and MEMS sensor behaves as a simple harmonic oscillator yielding the following equation:

$$\frac{d^2x}{dt^2} + 2\lambda\omega_o \frac{dx}{dt} + \omega_o^2 x = \frac{d^2u}{dt^2} \quad (7)$$

Where x is the plate position with respect to two surrounding electrodes (ex. above and below) and u is the electrodes position relative to an external inertial frame. Transformation to the s domain yields three equations which represent the electrical signals responses to the displacement, velocity and acceleration in a MEMS sensor:

$$H'_{displacement}(\omega) = A \frac{s^2}{s^2 + 2\lambda\omega_0 s + \omega_0^2} \quad (8)$$

$$H'_{velocity}(\omega) = A \frac{s}{s^2 + 2\lambda\omega_0 s + \omega_0^2} \quad (9)$$

$$H'_{acceleration}(\omega) = A \frac{1}{s^2 + 2\lambda\omega_0 s + \omega_0^2} \quad (10)$$

Using MATLAB again (see Appendix A), the frequency and phase step responses of the above transfer functions are obtained (see Figure 20, Figure 21 and Figure 22) with $\omega_0 = 10$ (rad/s) and $\lambda = 0.2$.

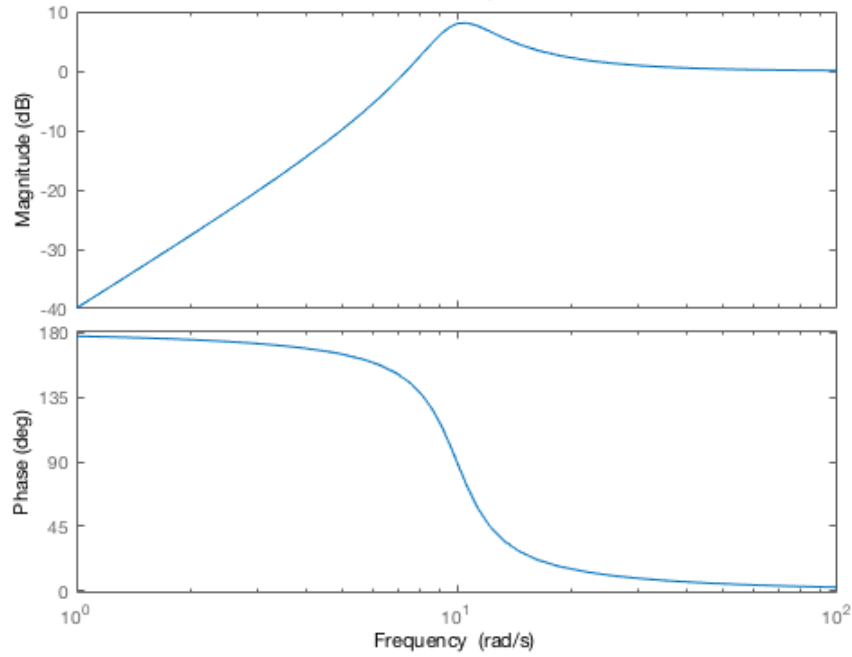


Figure 20. Displacement step response (normalized).

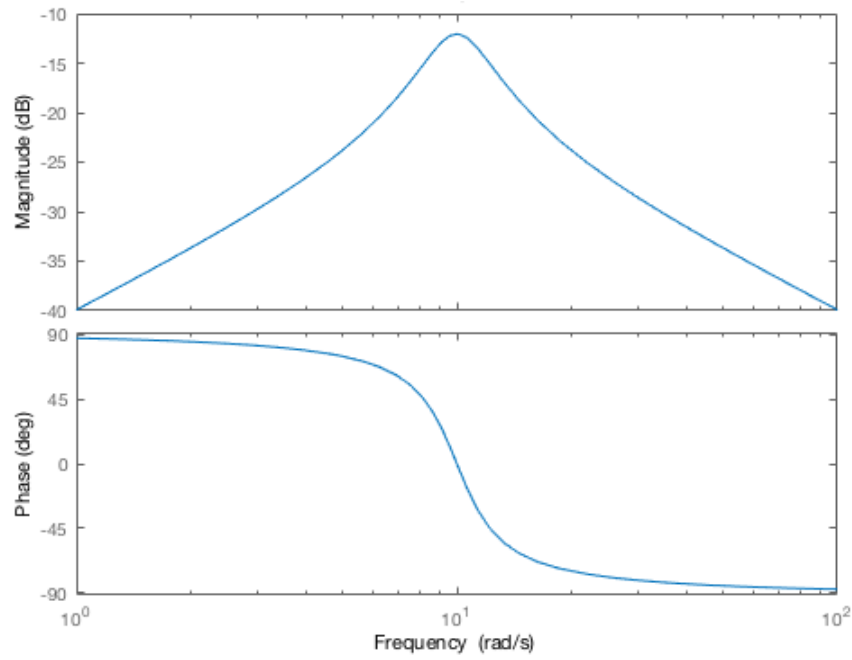


Figure 21. Velocity step response (normalized).

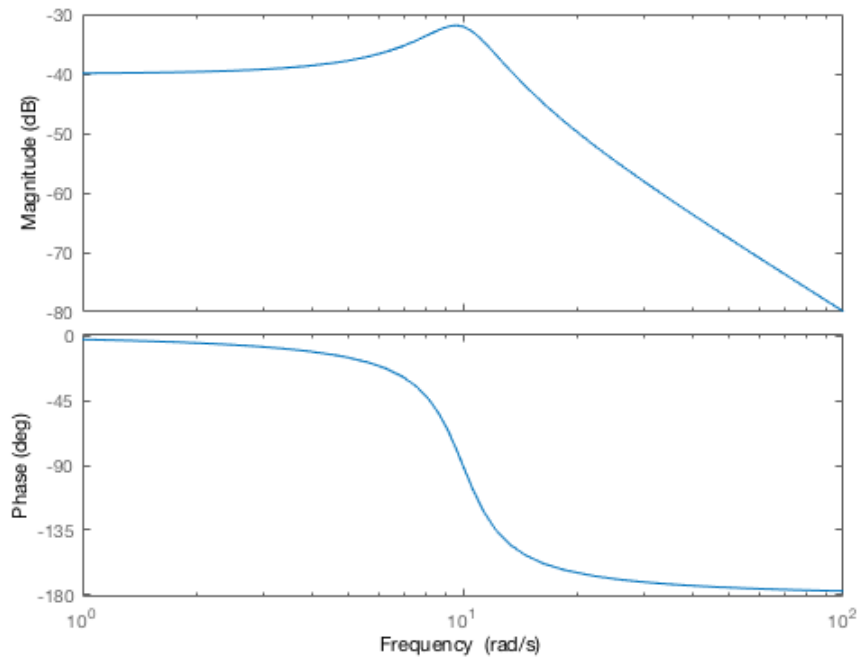


Figure 22. Acceleration step response (normalized).

As seen from the above figures, high pass, band pass and low pass responses can be achieved by taking the displacement, velocity and acceleration responses respectively. However, in the case of seismic data acquisition, the low pass behavior (along with the

almost zero phase) of the acceleration response is desirable and hence MEMS sensors commonly known as accelerometer. In [11], proper identifications of the undetermined constants were done to get a better modeling approximation.

3.3.2 Pre-ADC Conditioning

Given that the frequency of the seismic signal is in the range of 10mHz - 100Hz and amplitude of 1μV-1mV [18], the design will have the following specifications; inverting preamplifier with gain of 10, 4-kHz BW of the matching filter frequency range, 4 main gain options, clipper circuit, with diodes, and 24-bit ADC resolution including the digital anti-aliasing filter. The constructed signal conditioning circuit without the ADC is shown in Figure 23 [17]. The Appendix E shows the table of the components of the circuits and corresponding calculations for the gain and BW.

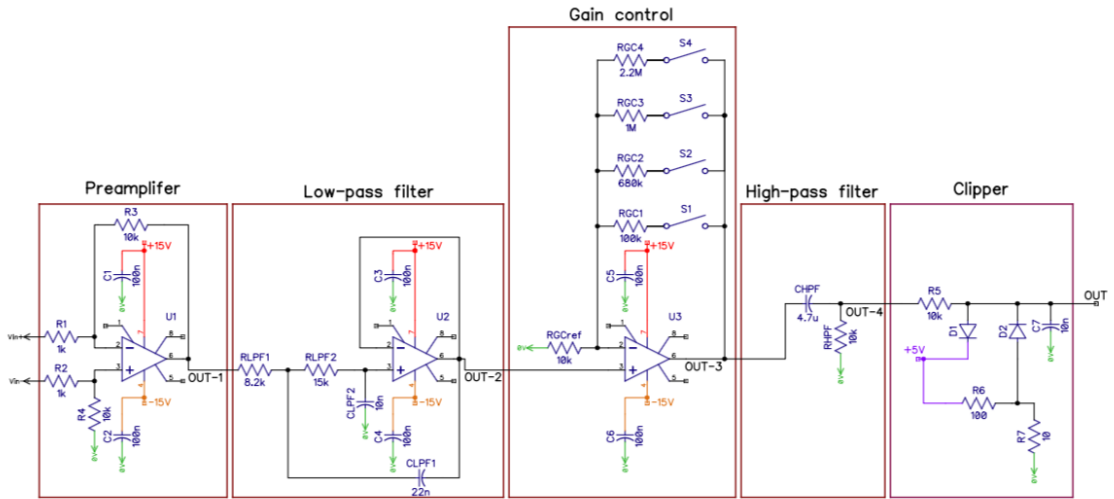


Figure 23. Pre-ADC conditioning circuits [17].

Preamplifier

The amplification is distributed into 2 stages, preamplifier and gain control, in order to reduce the referred noise. The first stage is the in the preamplifier as an inverting amplifier with differential source. The gain is shown in equation 11 which was set to 10.

$$G = -\frac{R_3}{R_1} \quad (11)$$

Low Pass Filter

Second order active low pass filter is designed with unity gain and 1kHz cut off frequency based on Sallen-Key topology. Equation 12 represents the LPF cut off frequency.

$$f_{LPF} = \frac{1}{2\pi\sqrt{R_{LPF1}R_{LPF2}C_{LPF1}C_{LPF2}}} \quad (12)$$

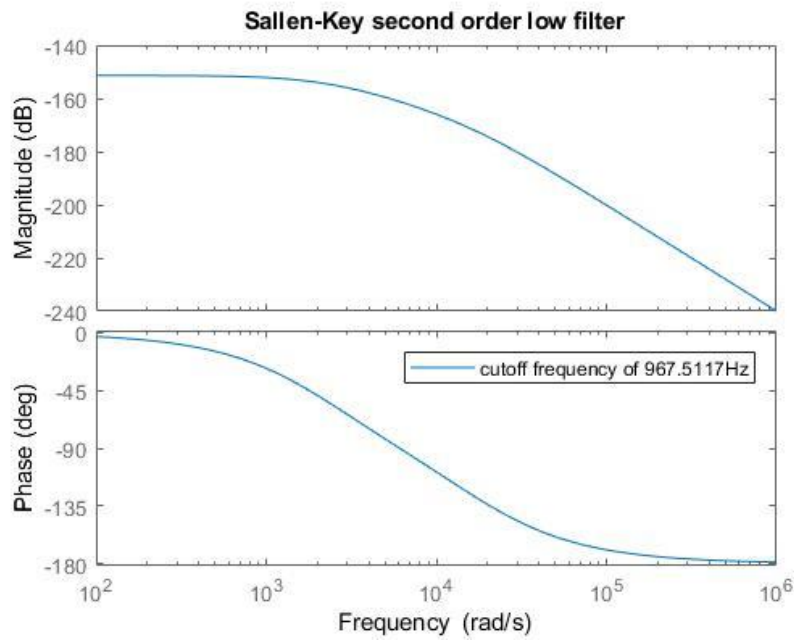


Figure 24: Frequency response of the active LPF

Figure 24 shows the ideal response of the LPF generated via MATLAB. The Q , ω_o and cut of frequency are 0.3223, $6.0791e+03$ rad and 967.5117 Hz respectively.

Gain Control Circuit

Gain control circuit is designed using non-inverting Op-Amp based amplifier integrated with basic resistor-based ladder topology. The gain equations are given by:

$$G_1 = 1 + \frac{R_{GC1}}{R_{GCref}} \quad (13)$$

$$G_2 = 1 + \frac{R_{GC2}}{R_{GCref}} \quad (14)$$

$$G_3 = 1 + \frac{R_{GC3}}{R_{GCref}} \quad (15)$$

$$G_4 = 1 + \frac{R_{GC4}}{R_{GCref}} \quad (16)$$

High Pass Filter

First order passive high pass filter is designed with 4Hz cut off frequency. Equation 17 represents the HPF cut off frequency:

$$f_{HPF} = \frac{1}{2\pi R_{HPF} C_{LPF}} \quad (17)$$

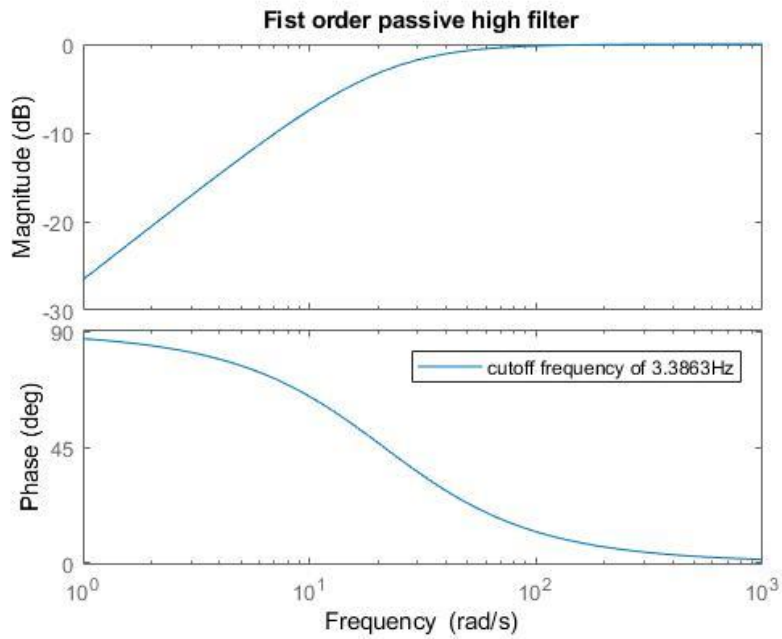


Figure 25. Frequency response of the passive HPF.

Figure 25 shows the ideal response of the HPF generated via MATLAB. The ω_o and cut of frequency are 21.2766 rad and 3.3863 Hz respectively.

Clipper

To ensure the safety of the ADC, clipper circuit based on normal diodes is used to ensure the chopping of voltages amplitude higher than the 2.5V that matches the maximum sensitivity detection of the ADC.

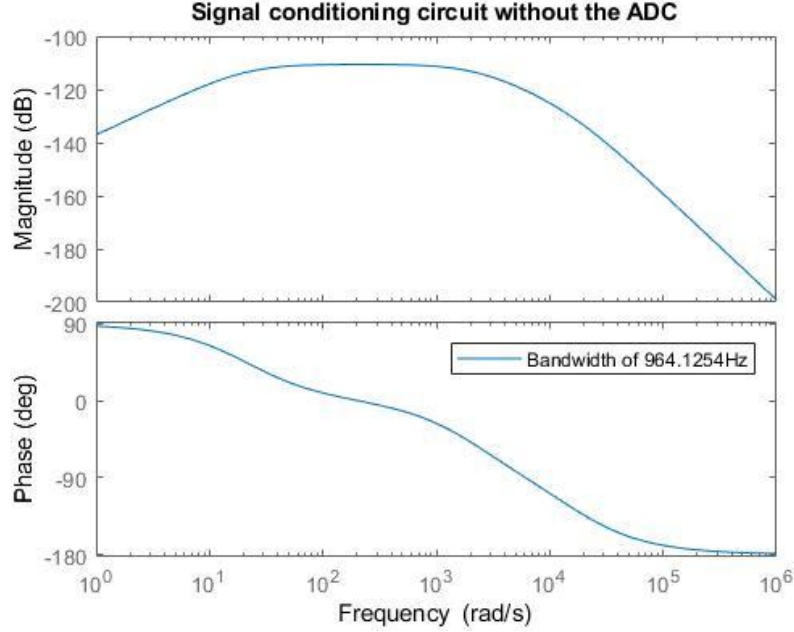


Figure 26: Frequency response of the S.C.C. without ADC

Figure 26 shows the ideal response of the S.C.C. without ADC generated via MATLAB. The gain and BW are 110 and 964.125 Hz respectively.

3.3.3 Wireless Communication

The seismic mechanical vibrator frequency is swept from about 2 Hz to 90 Hz. Knowing that the design uses 24 bits in ADC and data rate of ZigBee XBee is 250 kbps. Hence, theoretically the maximum number of geophones (MNoGs) signals that can be wirelessly transmitted is calculated as:

$$MNoGs = \frac{250kbps}{90Hz \times 2 \times 24bits} = 57 \text{ geophones} \quad (18)$$

Hence theoretically and ideal conditions, at single wireless node, Digi XBee RF module can handle up to 26 geophones at once. Practically it is lower than that.

Chapter 4

Final Design

Chapter 4 describes the final design of the project. An overall description (Section 4.1) starts before explaining the detailed description (Section 4.2) of the final implemented design. Analysis and results of the different elements are provided next (Section 4.3) along with cost analysis (Section 4.5). Finally engineering drawings will be shown (Section 4.6).

4.1 Detailed Design Description

The following sections will describe in detail each related element of the desired design project:

4.1.1 Sensors

Two Analog sensors and one digital sensor (see Figure 27) are used to observe the obtained seismic signals by single chock vibration source. The used sensors are: (1) Coil-Based Geophone SM-24 (analog), (2) Piezoelectric MiniSense 100 (analog) and (3) MEMS device GY-521 (digital). The analog sensors are connected to the constructed conditioning circuits before processing while the data produced by the digital sensor is ready for processing without conditioning.



Figure 27. Coil-Based Geophone (left), Piezoelectric (middle) and MEMS device (right).

4.1.2 Conditioning Circuits

For the ADC part, AD7714 chip [19] from ANALOG DEVICES were selected because it can be programmed to implement a Δ - Σ (24 bits) ADC with 2ms sampling interval (which is widely used for seismic data acquisition [12]). Moreover, it is especially fabricated using a CMOS technology that has a very low power dissipation (standby is around 15 μ W) along with an internal gain and low pass filter control. The chip pin configuration and functional block diagram are shown in Figure 28 and Figure 29 respectively.

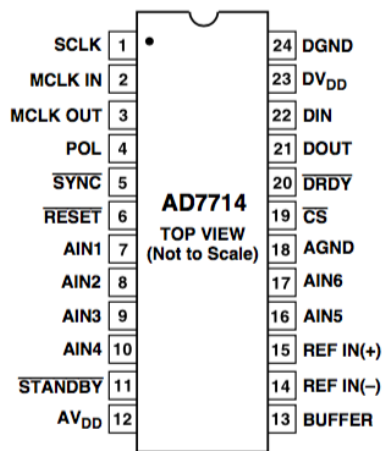


Figure 28. AD7714 Pin configuration.

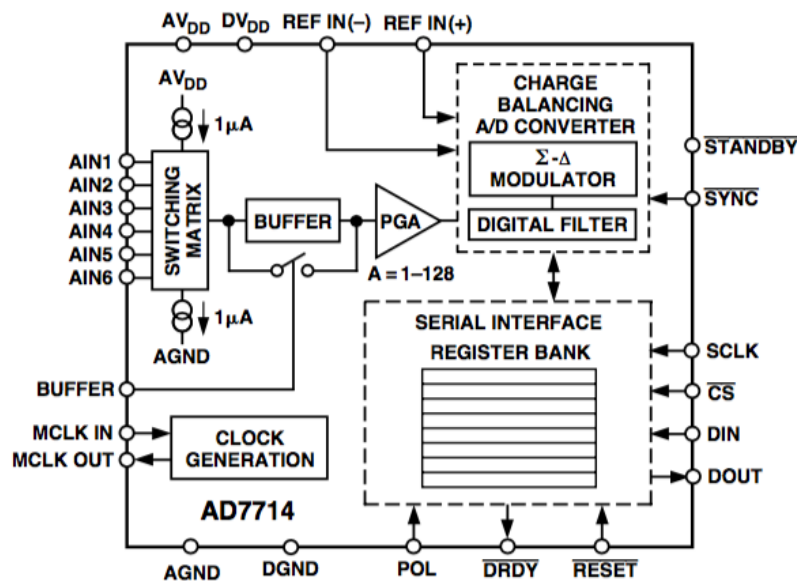


Figure 29. AD7714 functional block diagram.

The main task was to program the chip using Arduino Nano with SPI serial interfacing which was very difficult to achieve due to the lack of resources. However, after reading the datasheet and many extensive tests, verification of the correct operation lead to comprehensive understanding of several concepts such as: Nyquist sampling theorem, the concept of Δ - Σ conversion, SPI serial communication gain and filter programmability and the practical differences between single ended and differential input signals. 8 Registers within the chip were programmed to set the desired operation. Under the requirements of the final design, the AD7714 (ADC chip) was configured with the following settings:

- Fully differential input between AIN1 and AIN6 (see figure).
- Initial self-calibration mode when the chip is powered on (returns to the normal ADC conversion mode after a very short period of time).
- Sampling interval of 2ms.
- Zero gain (a suitable gain was used when the external conditioning circuits did not provide the required gain value).

The implemented circuit connections of the chip are shown in Figure 30 where a reference voltage supply (AD780) was used to achieve maximum precision.

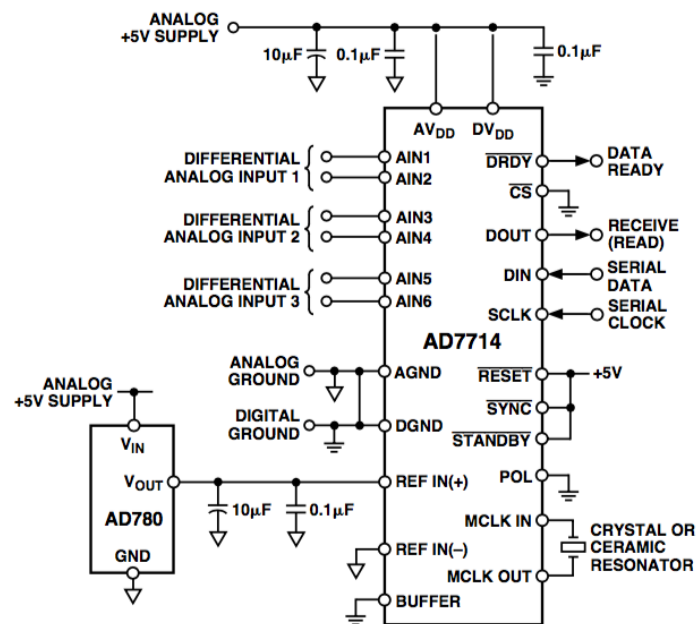


Figure 30. AD7714 circuit connections.

4.1.3 Signal Processing

To perform a way of taking the data from ADC and send it to central hub, the design needs a controlling unit that organizes and achieves this task. There are four suggested units for which some of them are proposed in the literature such as FPGA. The other three are microcontrollers as shown in Table 5. Geophone nodes are usually left to collect data for long time and thus power consumption is a crucial factor. Thus, most of those units have low power consumption. SPI interface is needed to be used to communicate with the ADC which requires 3 wires in addition to other 2 wires for chip enabling and selecting. ATtiny85 has less number of pins which is about 6 pins without the ground and power supply which is not enough. So, this choice is rejected. Nexys3 needs to be programmed using HDL which is somehow complicated compared to C language. Thus, AT89s8253 and ATmega328P are preferred. Since ATmega328P comes with a well prepared integrated board, it makes it simple to program and reprogram for any future modification. Thus, Arduino nano with ATmega328P is chosen.

Table 5. Comparison between the available microcontrollers.

Feature	FPGA	Microcontrollers		
	Nexys3	ATtiny85	AT89s8253	ATmega328P
Number of Pins	324	8	40	28
Memory Size (Bytes)	16M	8K	12K	32K
Chip Size	-	10.1mm x 7.11mm	52mm x 13.5mm	43.18mm x 18.5mm
Power Consumption	3.3V 200mA	-	5.5V 10mA	5 V 10mA

Feature	FPGA	Microcontrollers		
	Nexys3	ATtiny85	AT89s8253	ATmega328P
Programming Method Complexity	HDL	Arduino IDE C Language	C Language	Arduino IDE C Language

4.1.4 Wireless Communication

RF Module

For its large line-of-sight transmission range, compared to nRF24L01+ module, Xbee S2C Zigbee was chosen over nRF24L01+. This RF module uses Zigbee standard that is based on IEEE 802.15.4 standard for low-power, low-cost and wireless mesh-networking. Xbee could be configured as a coordinator, router or an end-device. In each Personal Area Network (PAN) there must be one and only one coordinator. The coordinator is a Zigbee node that performs the unique function of forming the network. It also has similar functions to the router node. Router node uses network information to find the best route for data packets. The last Zigbee node is an end-device node, it only receives and transmits data and cannot route them. One coordinator, at least one router and, optionally, one or more end-devices are needed in each PAN. For their essential tasks, coordinators and routers cannot be set to sleep mode, on contrast, an end-device allows sleep mode. After configuring the device, its operating mode should be set up. Xbee S2C offers application programming interface (API) and Transparent operating modes. Which mode to select depends on needed application. For example, when the application uses a large network, API mode is recommended for its ease of data transmission management to multiple destination. If the application doesn't need such complexity, such as our design, transparent mode is recommended for its simple interface. Another factor to decide which mode to use is whether the application needs a UART or SPI serial interface. If 250kbps, or less, is needed then Xbee can be used with transparent mode using UART interface. But if data rate up to 5Mbps was needed

then user must use SPI interface and operate it with API mode (Transparent is not available for SPI).

This design used only coordinator and end-device nodes with same PAN and used transparent operating mode. Future upgrading of the design will consist of more routers and end-devices. Figure 31 shows a generic Zigbee network. Readers may refer to figure 25, in Chapter2, which presents the RF module used in the project (Xbee S2C Zigbee RF module).

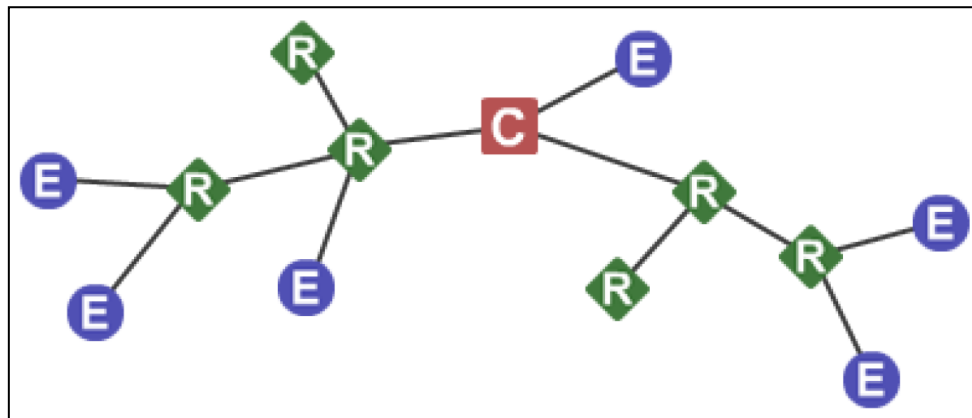


Figure 31. A generic Zigbee Network, where: C, R and E stands for Coordinator, Router and End-Device respectively.

4.1.5 Storage and Processing Unit

The design offers two options for the user. One is to make the receiving point either storage or a processing unit (or even both at the same time). An SD card module is connected to receiver's controller and stores its reconstructed signal. When seismic survey ends, user recollects the stored data from the SD card and transfers it to the processing unit to extract information. User has the other option to only process the data and reconstruct information. It is also possible to process and store data at the same time (this is the default settings of the design).

4.2 Analysis and Results

Several experiments were done to ensure the reliability of the operation of the individual components. The obtained results are listed (Section 4.3.1) and analyzed (Section 4.3.2) correspondingly.

4.2.1 Results of Pre-ADC conditioning Circuits

Using analog discovery 2, the S.C.C. without ADC is tested and verified as the shown below.

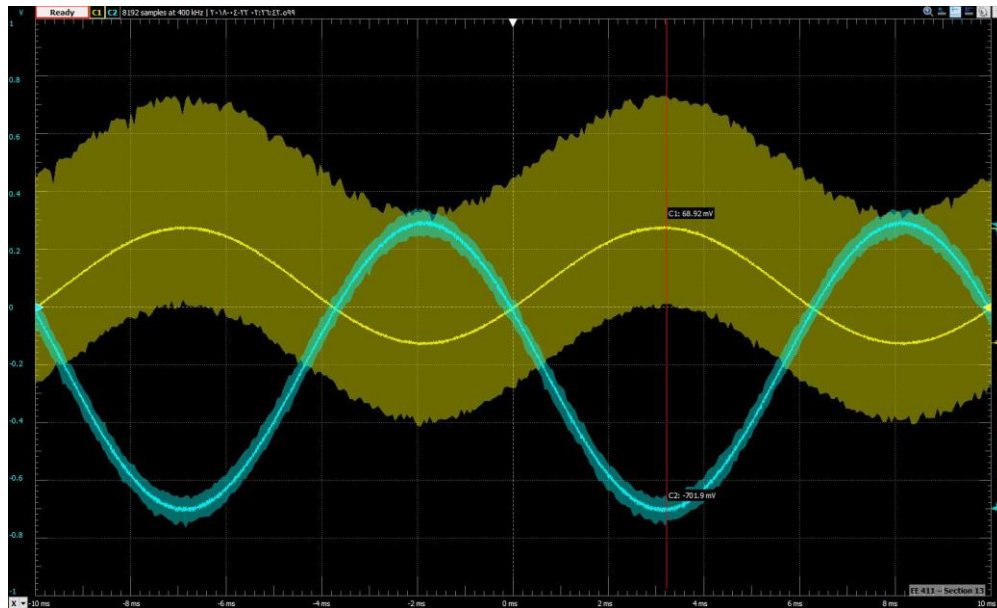


Figure 32: Preamplifier result

As shown in Figure 32, the amplification of preamplifier stage is measured to be 9.946 out of phase with source signal. The calculated gain is 9.939. Hence, this stage is verified.

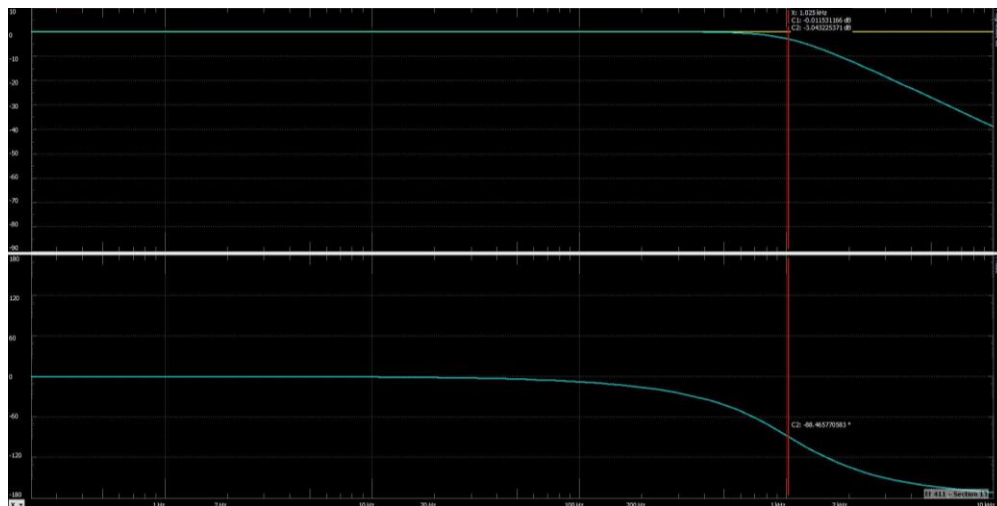


Figure 33: LPF response

As shown in Figure 33, the cut of frequency of the LPF stage is measured to be 1.025 kHz. The calculated cut of frequency is 966.57 Hz. Hence, this stage is verified.

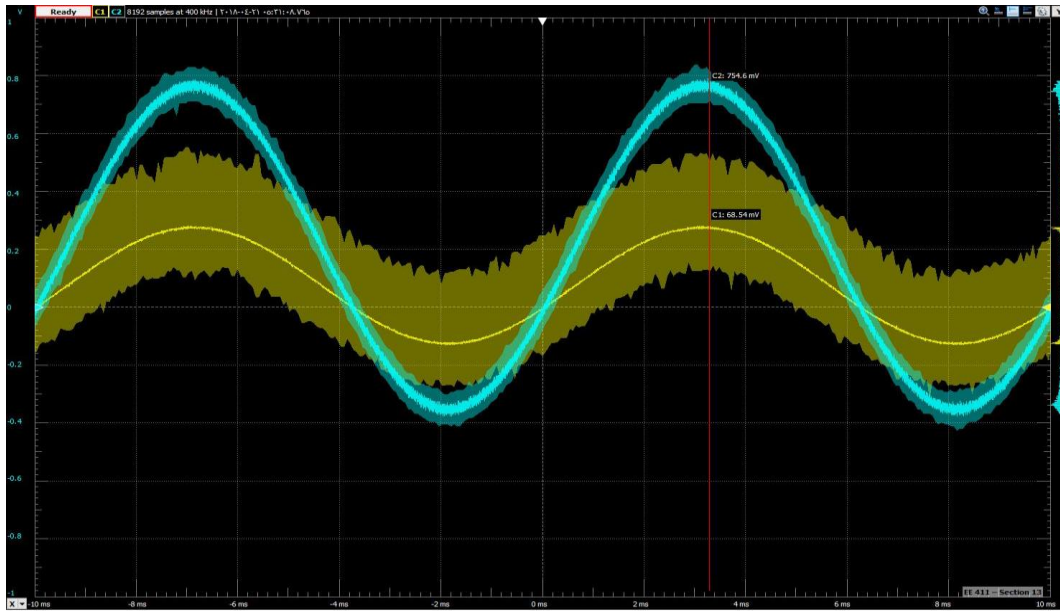


Figure 34: Gain control option 1 result

As shown in Figure 34, the amplification of Gain control option 1 stage is measured to be 10.847 in phase with source signal. The calculated gain is 10.163. Hence, this stage is verified.

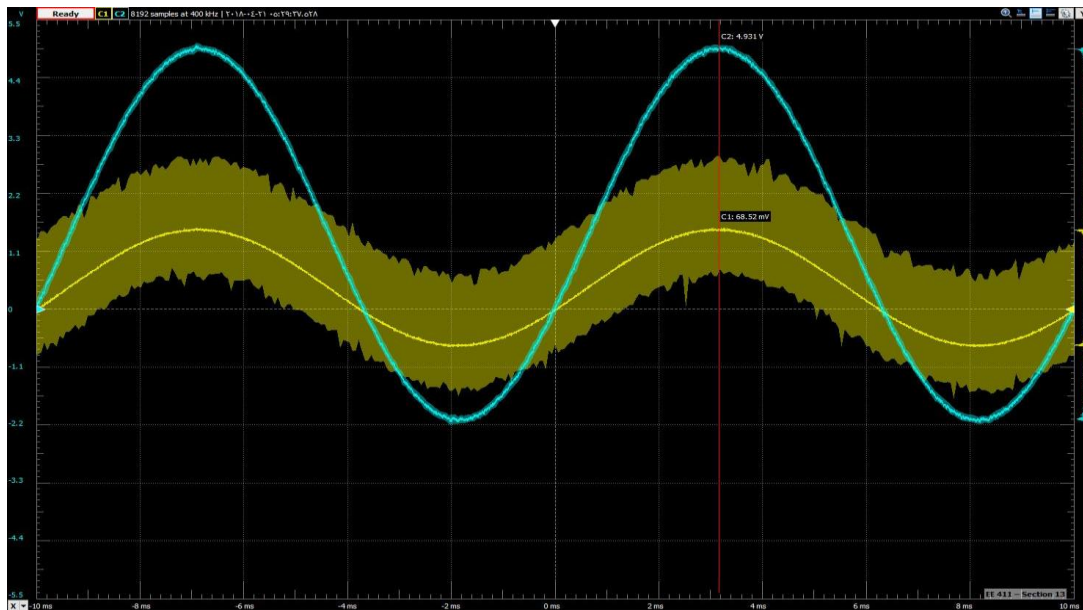


Figure 35: Gain control option 2 result

As shown in Figure 35, the amplification of Gain control option 2 stage is measured to be 70.653 in phase with source signal. The calculated gain is 69.694. Hence, this stage is verified.

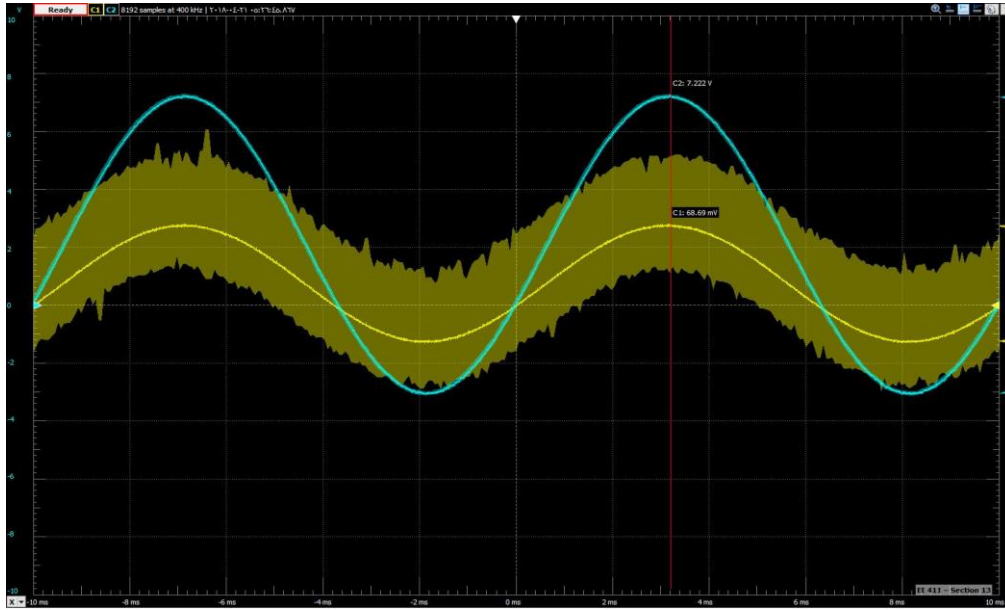


Figure 36: Gain control option 3 result

As shown in Figure 36, the amplification of Gain control option 3 stage is measured to be 102.895 in phase with source signal. The calculated gain is 101.837. Hence, this stage is verified.

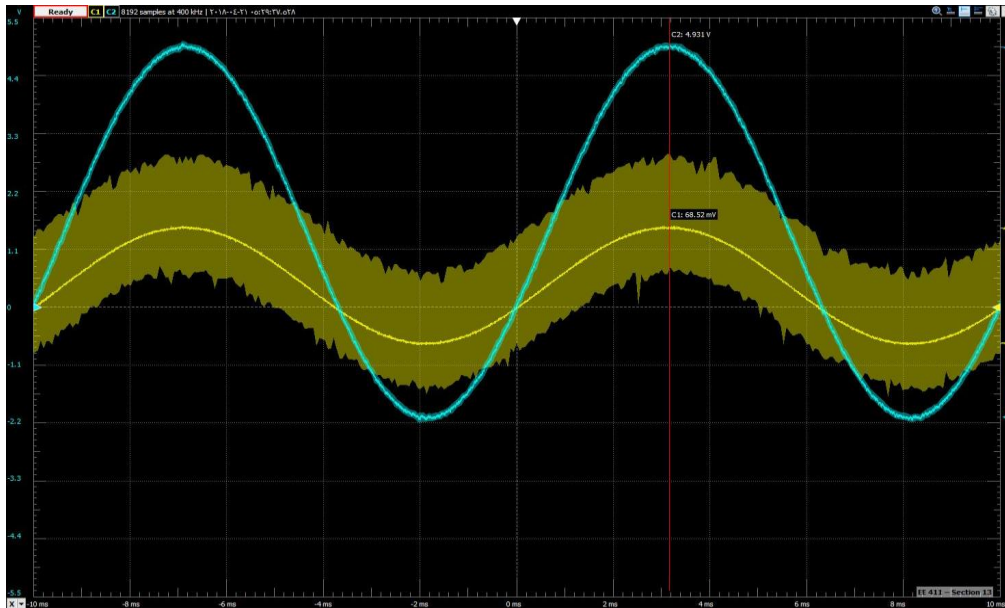


Figure 37: Gain control option 4 result

As shown in Figure 37, the amplification of Gain control option 4 stage is measured to be 205.326 in phase with source signal. The calculated gain is 203.571. Hence, this stage is verified.

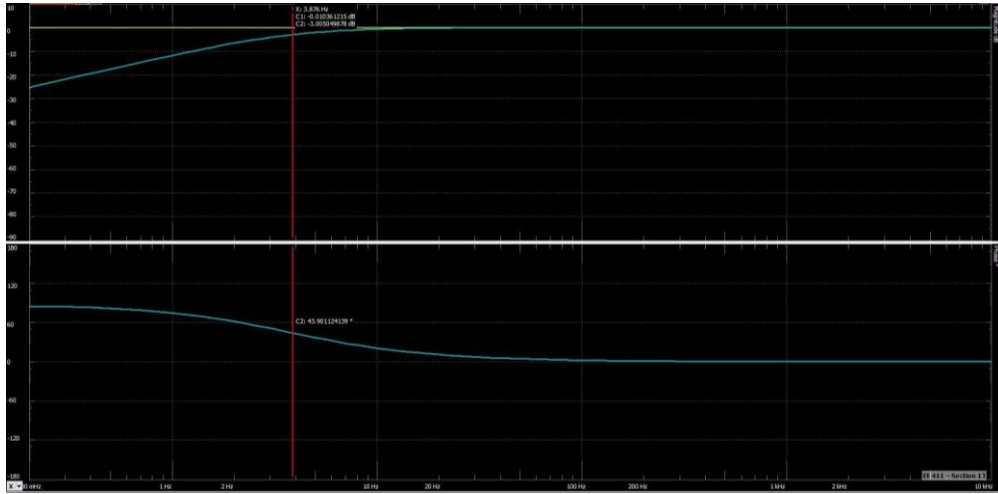


Figure 38: HPF response

As shown in Figure 38, the cut of frequency of the HPF stage is measured to be 3.876 Hz. The calculated cut of frequency is 3.47 Hz. Hence, this stage is verified.

Finally, according to the results shown above for LPF and HPF stages, the BW of the S.C.C without the ADC is measured to be 1021.12 Hz. The calculated BW is 963.11 Hz. Whereas the minimum gain and the maximum gains are 101.0129 and 2023.301 respectively. The source signal applied is of 50 mV amplitude and 100 Hz frequency.

4.2.2 Analysis of Sensors with ADC

The impulse response of the three sensors is achieved by a very quick and strong hummer strike near the sensor. Three results of are obtained for the (1) Coil-Based Geophone (analog), (2) Piezoelectric (analog) and (3) MEMS device (digital) as shown, in section4.2.2, in Figure 39, Figure 40 and Figure 41 respectively.

4.2.3 Results of Sensors with ADC

The Coil-Based Geophone (Figure 39) shows a smooth transition when compared with the other two and the highest observed peak reached around 800mV. A highly oscillatory behavior is observed on the Piezoelectric sensor (Figure 40) with a peak value exceeding 1250mV. The accelerometer (MEMS) response (Figure 41) have fast settling time with few oscillations. All of them appear to have an underdamped response. The

absolute time scale is not important in these cases (and later ones) because the objective here is to observe the overall quality of the obtained signals.

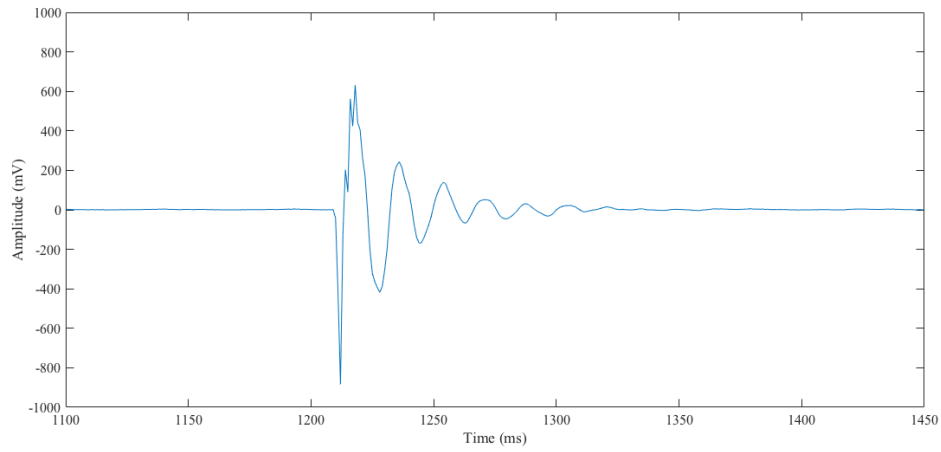


Figure 39. Coil-Based Geophone impulse response.

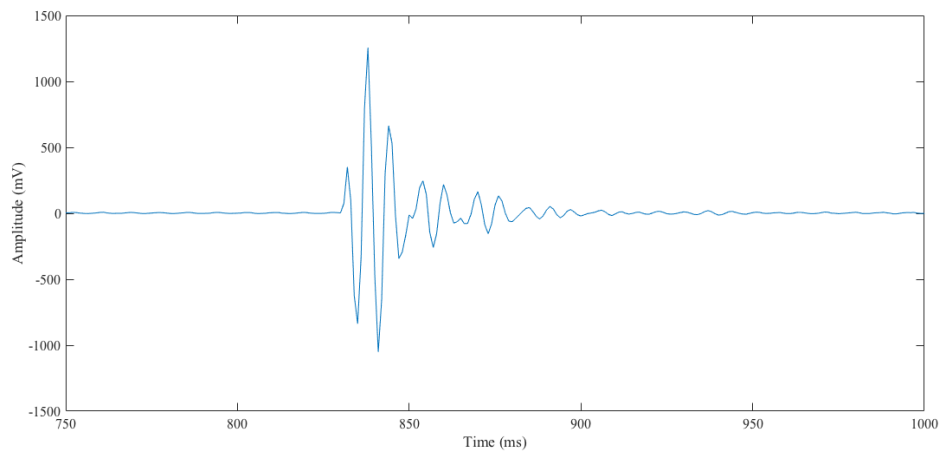


Figure 40. Piezoelectric impulse response.

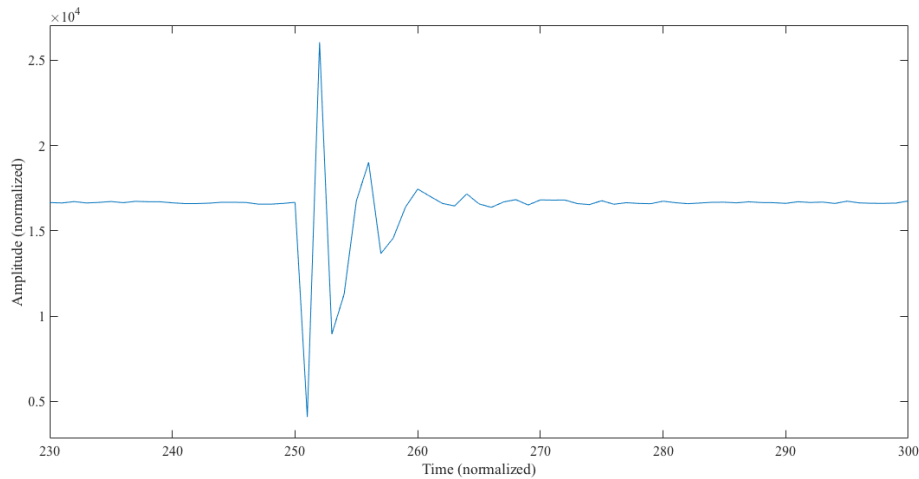


Figure 41. Accelerometer impulse response.

4.2.4 Analysis of Wireless Communication

One of the ways to make a seismic survey is by generating seismic waves with frequency that is swept from 2Hz to 90Hz. Hence, a signal generator was used to generate a sweep of frequencies that starts at 2Hz and ends at 90Hz. The single tone sinusoidal signal was fed to the ADC to convert it to a digital signal of 24 bits. After ADC stage, data were sent wirelessly from ADC to the receiving point to collect them. These data were then transferred to MATLAB to plot their Fast Fourier Transform (FFT) and find the frequencies of the peaks. As expected, at low frequencies there was only two peaks (negative and positive Fourier frequencies) and negligible noise peaks, however, as the signal frequency increases the noise peaks draw more attention to the observer.

To reinsure the reliability of the test, a MATLAB elliptic Band Pass Filter (BPF) was designed with different bandpass ranges, each range was chosen for a certain frequency. All collected signals were passed through this filter then were applied to FFT to find the frequency of maximum peaks. Results (peaks' frequencies), with and without the BPF, matched to each other. Results are introduced in the next section (4.3.2). Finally, used MATLAB code can be reviewed in Appendix B and plot of one of the used BPFs is shown in Figure 42.

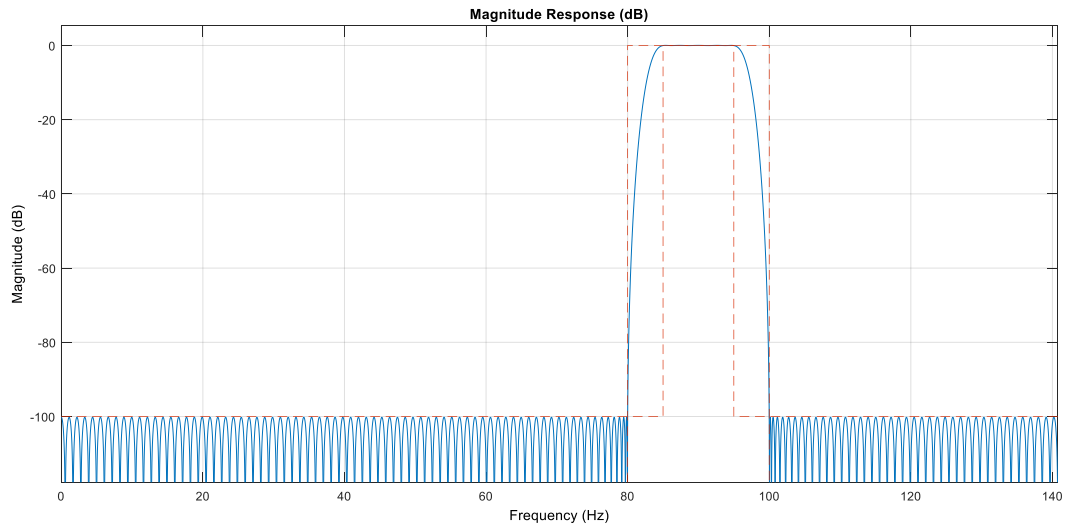


Figure 42. First stopband frequency = 80Hz, first passband frequency = 85Hz, second stopband frequency = 95Hz, second passband frequency = 100Hz, stopband attenuation = -100dB and passband ripples variation = 0.05dB.

4.2.5 Results of Wireless Communication

The essential results of the wireless communication test are summarized on **Table 6**. Reader can refer to Appendix C to review FFT plots of all signals with peak information shown on them. Please refer to section 4.2.4 for more details on the conducted experiment which concluded these results.

Table 6. Summary of the final results.

Wirelessly sent frequency	Corresponding FFT peak frequency	Wirelessly sent frequency	Corresponding FFT peak frequency
2 Hz	2.0 Hz	35 Hz	35.2 Hz
5 Hz	5.0 Hz	50 Hz	50.1 Hz
8 Hz	8.1 Hz	60 Hz	60.2 Hz
10 Hz	10.0 Hz	70 Hz	70.3 Hz
15 Hz	15.1 Hz	80 Hz	80.2 Hz

Wirelessly sent frequency	Corresponding FFT peak frequency	Wirelessly sent frequency	Corresponding FFT peak frequency
20 Hz	20.1 Hz	90 Hz	90.3 Hz
25 Hz	25.2 Hz	---	---

For the sake of comparison, Figure 43 shows FFT plot of the 90Hz signal before and after applying the BPF (that is shown in Figure 42) to it.

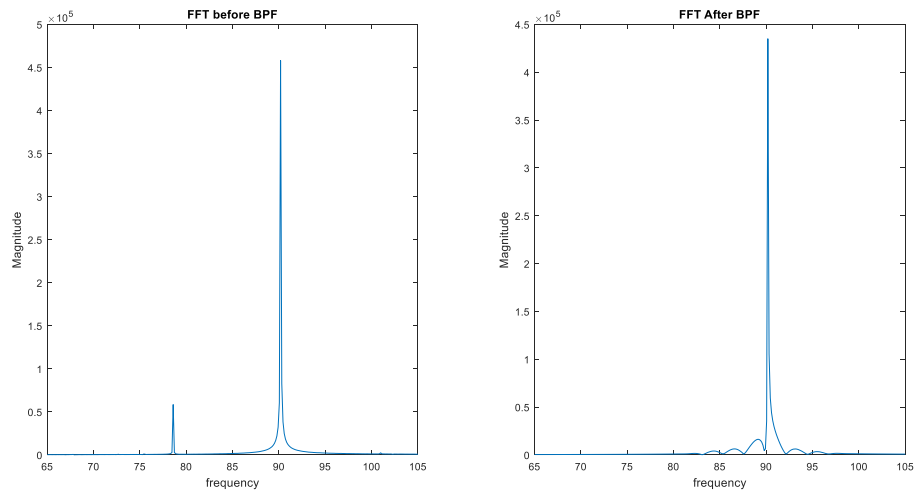


Figure 43. FFT plot of the 90Hz signal before and after applying the BPF.

4.3 Cost Analysis

Table 7. Complete list of the bought parts.

No.	Part Number	Quantity	Unit Price (USD)	Extended Price (USD)
1	ATTINY85-20PU	3	1.48	4.44
2	AT89LP213-20PU	3	1.39	4.17
3	1N4001-G	3	0.11	0.34
4	AD7714YNZ	1	19.73	19.73

No.	Part Number	Quantity	Unit Price (USD)	Extended Price (USD)
5	AT89S8253-24PU	3	3.28	9.84
6	XB24CZ7WIT-004	2	18.24	36.48
7	TLC0820ACN	1	5.88	5.88
8	TC7109CPL	1	8.94	8.94
9	LDT0-028K/L	2	3.84	7.68
10	1006015-1	2	4.19	8.38
11	VXO7803-500	2	2.39	4.78
12	VXO7805-500	2	2.39	4.78
13	VXO78012-500	2	2.39	4.78
14	NE5532P	2	1.04	2.08
15	OP07CP	6	0.90	5.42
16	AD620ANZ	2	10.31	20.62
17	OPA2705PA	2	2.45	4.90
18	OPA277PA	3	3.45	10.35
19	OPA27GP	3	3.59	10.77
20	CB10LV104M	3	0.40	1.20
21	P120K-Y20B100K	3	0.68	2.03
22	1N5231BTR	2	0.13	0.27
23	NMJ0515SC	1	14.39	14.39
24	NMJ0505SC	1	14.39	14.39
25	LFXTAL003185BULK	5	0.22	1.08

No.	Part Number	Quantity	Unit Price (USD)	Extended Price (USD)
26	22R224C	3	0.67	2.01
27	22R223C	3	0.67	2.01
28	AD780ANZ	1	11.60	11.60
29	FEMALE HEADERS	2	1.06	2.12
30	MALE HEADERS	2	1.06	2.12
31	XBEE TO USB ADAPTER	2	8.00	16.00
32	Accelerometer	1	6.66	6.66
33	Battery Holder	2	1.60	3.20
34	Load Cell Sensor	1	2.66	2.66
35	Triple Axis Accelerometer	1	4.00	4.00
36	Voltage Regulator	1	0.80	0.80
37	RF Transceiver	2	5.33	10.66
Total				271.55\$ = 1018.32SR
Total used in the final prototype				105.51\$ = 395.66 SR

Chapter 5

Prototype Implementation

This chapter will present the steps followed to achieve the final prototype in Section 5.1. Comparison between the prototype design with the planned design will be considered in Section 5.2 and highlights on the future manufacturing recommendations for possible mass production execution are discussed in Section 5.3.

5.1 Implementation Process

The main parts of the prototype are: (1) Sensor, (2) Conditioning Circuits, (3) Wireless Transmitter, (4) Wireless Receiver, (5) Microcontroller and (6) Emulated Seismic Environment Case. Moreover, different sensors are considered in the design: Coil-Based, MEMS and Piezoelectric. A signal condition circuit is used to purify the signal received by ADC and eliminate the noise consisting of a low pass filter, gain controller and a high pass filter. High resolution 24-bits Δ - Σ ADC is used in order to convert the analog seismic signal into digital words. The microcontroller in the design uses SPI Serial Communication to manage the command operations of the Δ - Σ ADC and the Wireless Communication (two separate microcontrollers are used: one at the transmitter side and another at the receiver side). Finally, the Emulated Seismic Environment Case was constructed on a rectangular plastic box with dimensions of (43x33x26) cm³ which will be filled with approximate realistic seismic layers as shown in Figure 44.

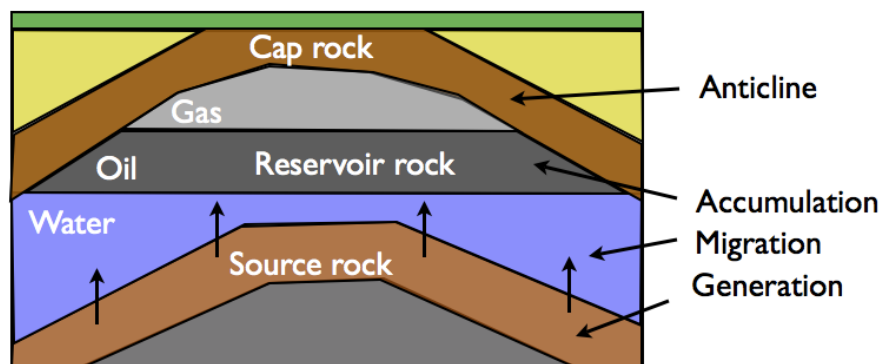


Figure 44. Upper earth crust layers.

5.2 Prototype vs. Planned Design

The implemented prototype was according to the plan except the use of 8051 microcontroller and FPGA due to their complicity when applied in such complex system with the time limitation (one semester). Also, the circuits were planned to be gathered on separate two Printed Circuit Boards (PCBs) by utilizing the PCB chemical lab at KFUPM (the designs are presented in Appendix H). However, due to several technical issues, we had to solder all the parts and connect every wire manually on different kind of boards.

Chapter 6

Design Verification

The final prototype was tested and verified experimentally with the help of the Geophysics Department at KFUPM under the supervision of Mr. Ayman Al-Lehyani and Mr. Abdul Latif Ashadi. Description of the experiment (Section 6.1) is followed by detailed results (Section 6.2) comparing the recorded data from their system and our system simultaneously.

6.1 Verification Test Description

The experimental test was made in a field trip near building 26 at KFUPM. A Coil-Based Geophone was connected in the implemented prototype in parallel with Mr. Abdul Latif recording seismic system before being embedded into the ground. A hammer was used to strike the surface of the earth at a short distance from the embedded sensor. When the generated seismic waves are reflected from the subsurface layers, a seismic signal are recorded from the sensor. The results are shown in following section (Section 6.2) as figures with captions indicating the shot number and self-explanatory experimental data used. Only the signals of interest are gathered where the sensor transforms the seismic waves into electrical signals. Mr. Abdul Latif system consist of a recording cabled seismic recording system which reads the same signal as the one we are reading (from a single sensor) and in all cases (Figure 45, Figure 48, Figure 50, Figure 52, Figure 54, Figure 56 and Figure 58) the obtained data shows clear similarities with the Geophysics system (Figure 45, Figure 47, Figure 49, Figure 51, Figure 53, Figure 55 and Figure 57 respectively) except when the Pre-ADC conditioning circuits are connected. These issues were related to unappropriated resultant gain that magnified the background noise and distorted the signals as shown in Figure 60 and Figure 62 which should compare with Figure 59 and Figure 61 respectively. The complete Arduino code of which the final prototype was programmed with is shown in Appendix C.

6.2 Detailed Results



Figure 45. Shot 1 from the Geophysics department system.

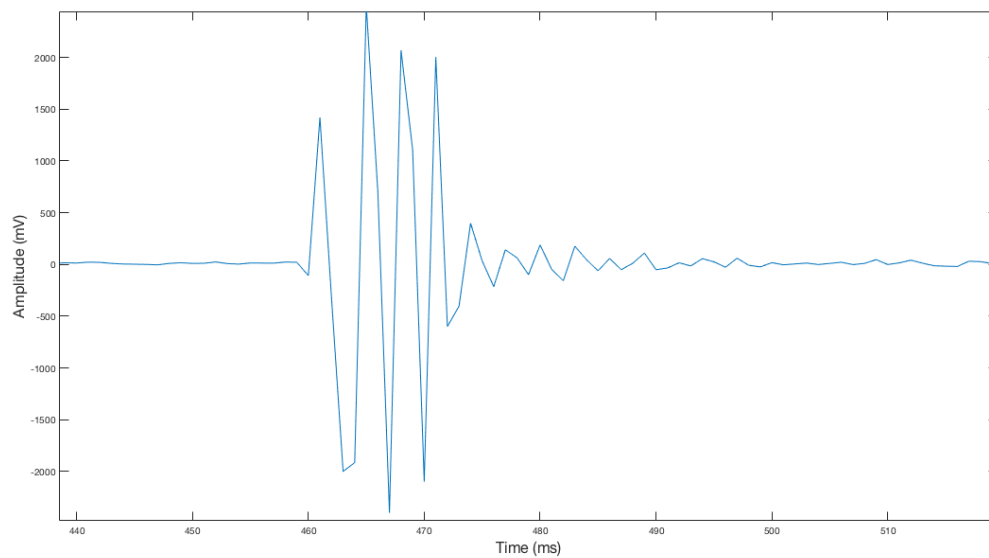


Figure 46. Shot 1 from our system (with ADC gain) (without pre-ADC conditioning).



Figure 47. Shot 2 from the Geophysics department system.

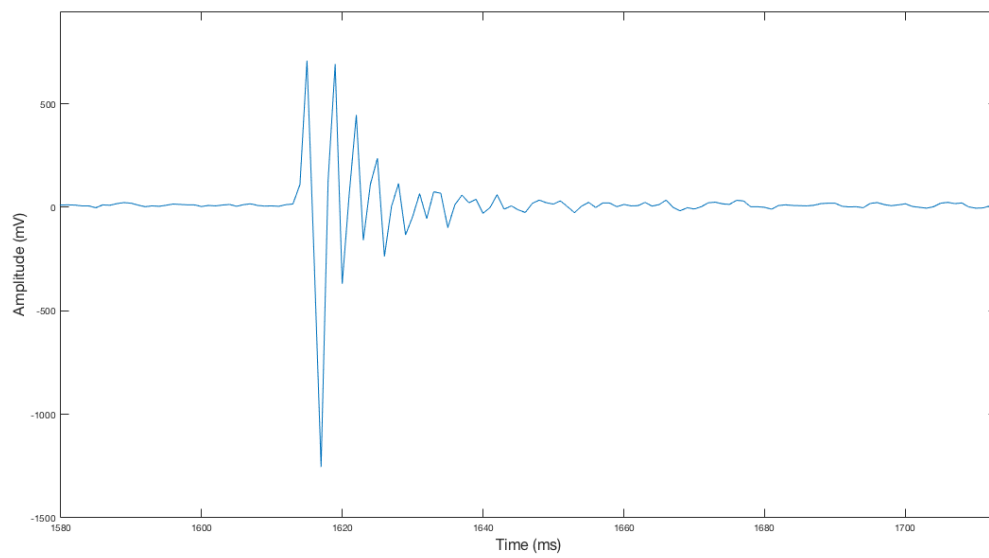


Figure 48. Shot 2 from our system (without ADC gain) (without pre-ADC conditioning).



Figure 49. Shot 3 from the Geophysics department system.

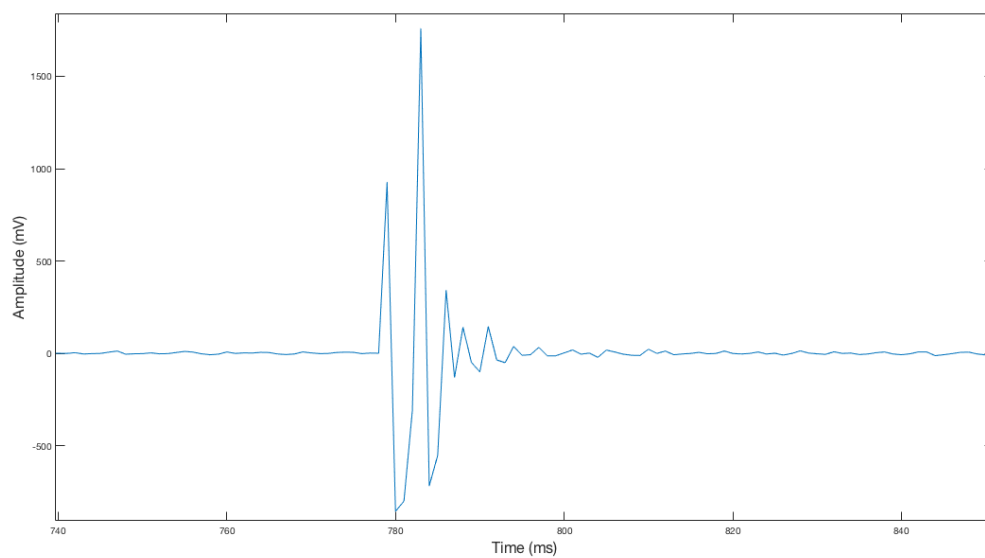


Figure 50. Shot 3 wirelessly received with our system (with ADC gain) (without pre-ADC conditioning) (0m).

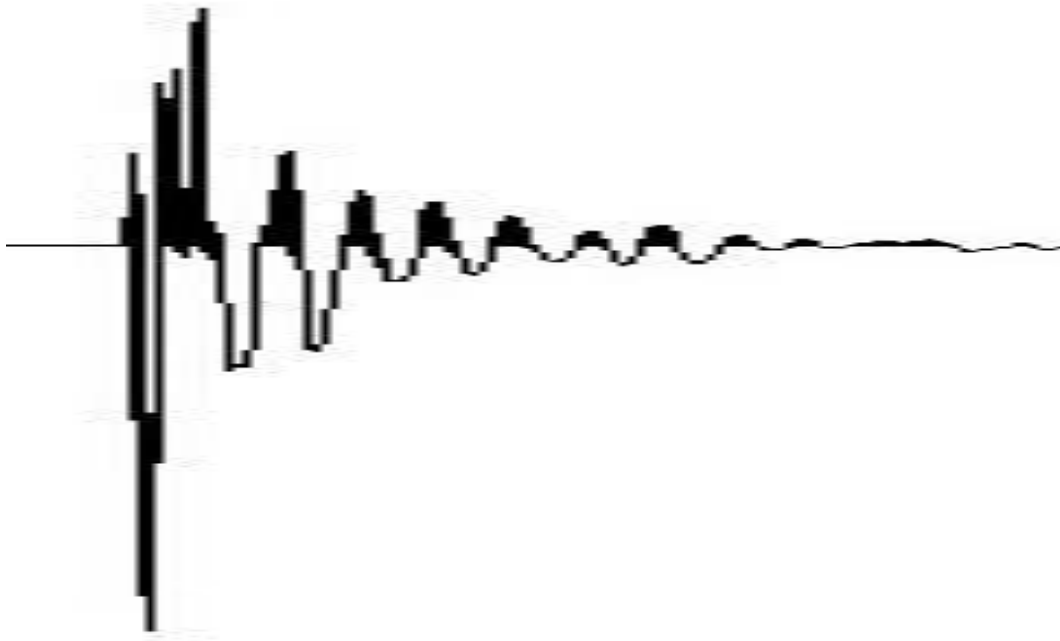


Figure 51. Shot 4 from the Geophysics department system.

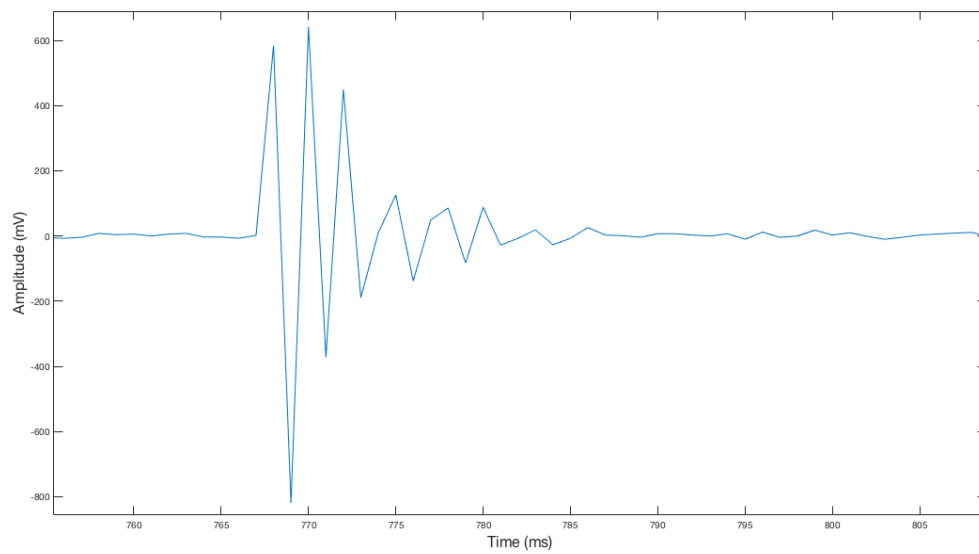


Figure 52. Shot 4 wirelessly received with our system (with ADC gain) (without pre-ADC conditioning) (10m).

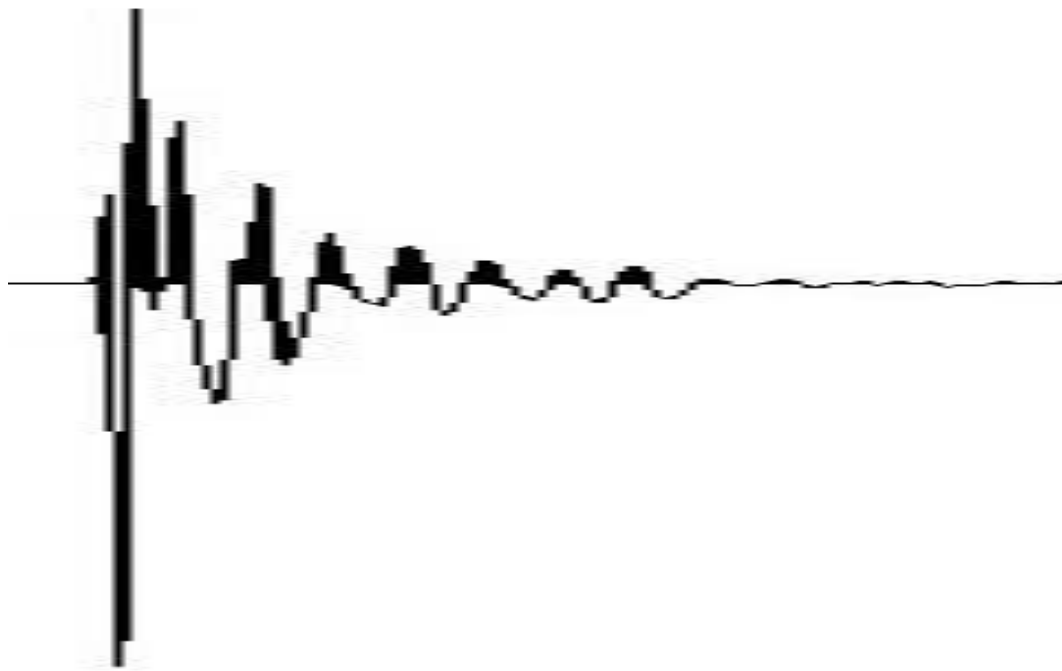


Figure 53. Shot 5 from the Geophysics department system.

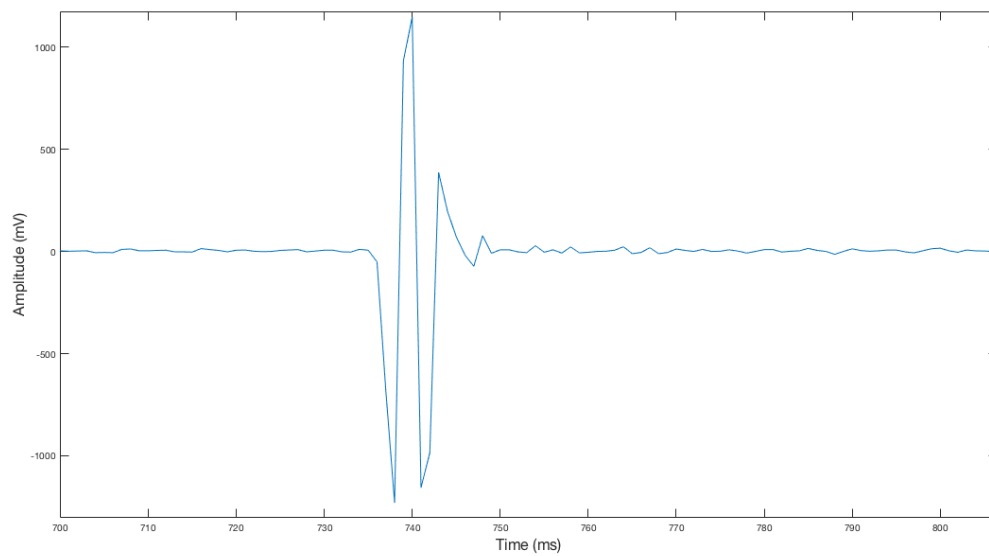


Figure 54. Shot 5 wirelessly received with our system (with ADC gain) (without pre-ADC conditioning) (5m).

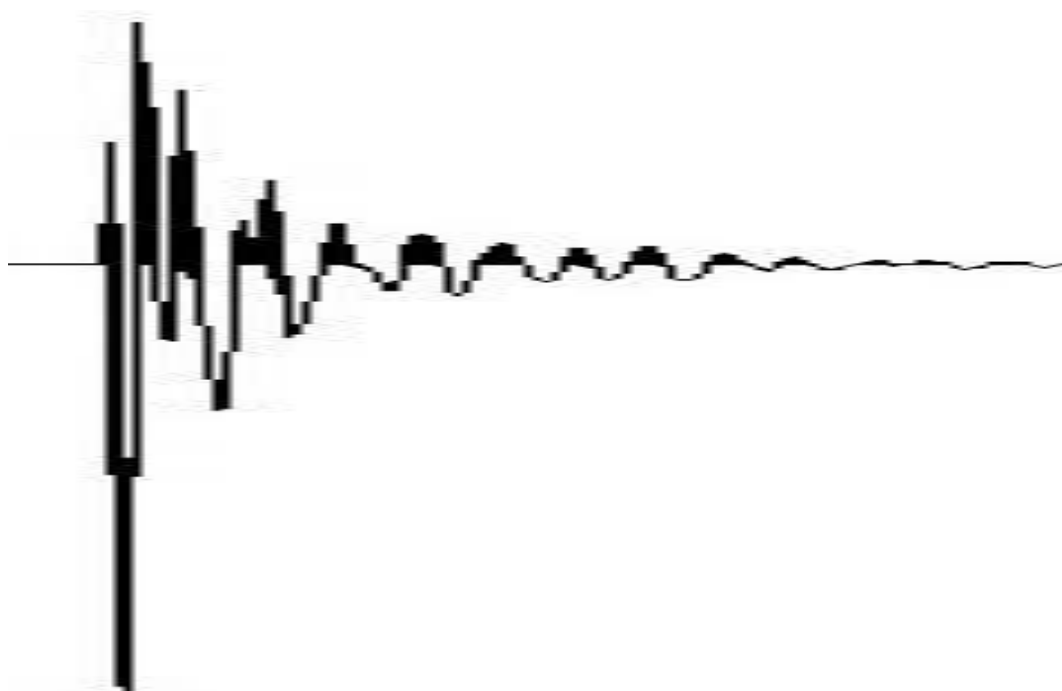


Figure 55. Shot 6 from the Geophysics department system.

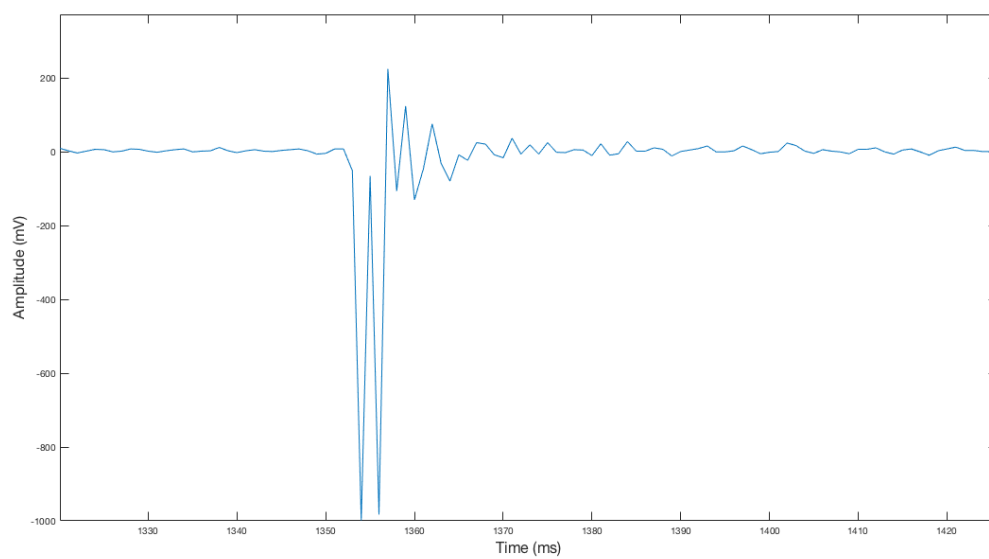


Figure 56. Shot 6 wirelessly received with our system (with ADC gain) (without pre-ADC conditioning) (50m).

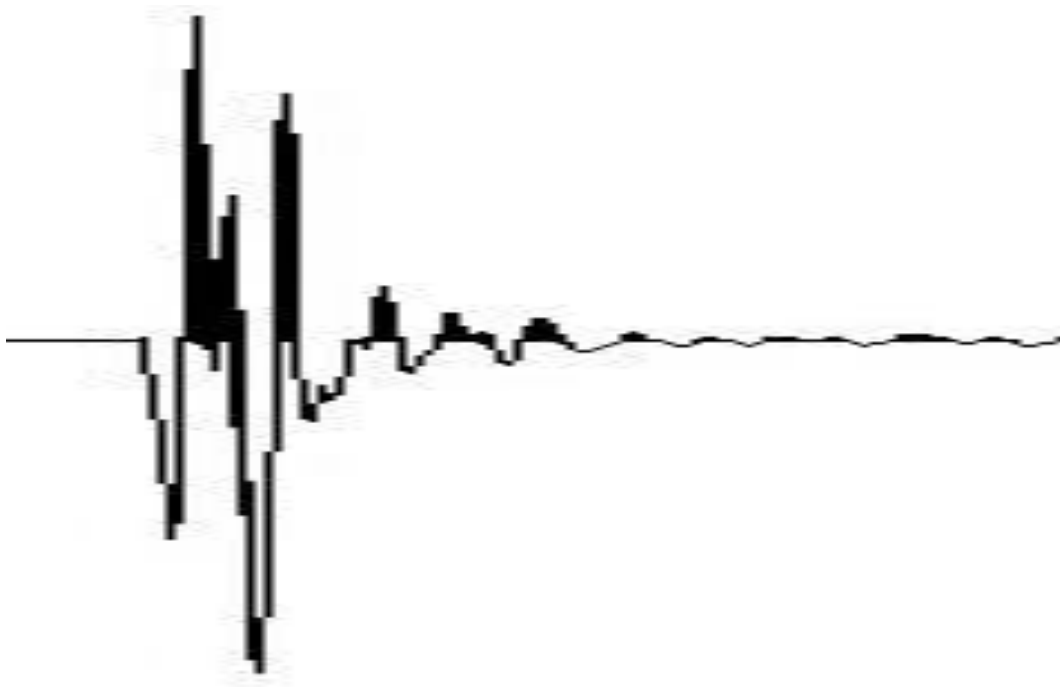


Figure 57. Shot 7 from the Geophysics department system.

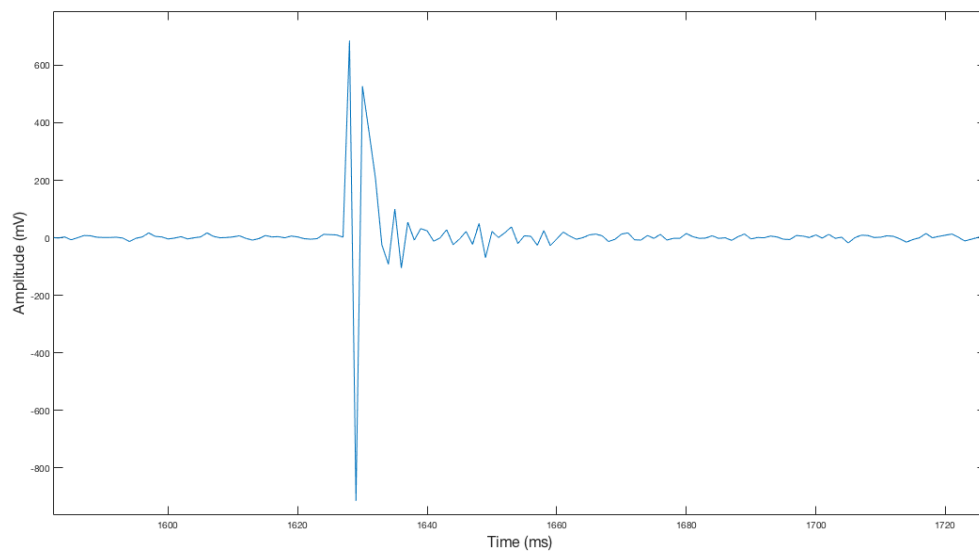


Figure 58. Shot 7 wirelessly received with our system (with ADC gain) (without pre-ADC conditioning) (25m).



Figure 59. Shot 8 from the Geophysics department system.

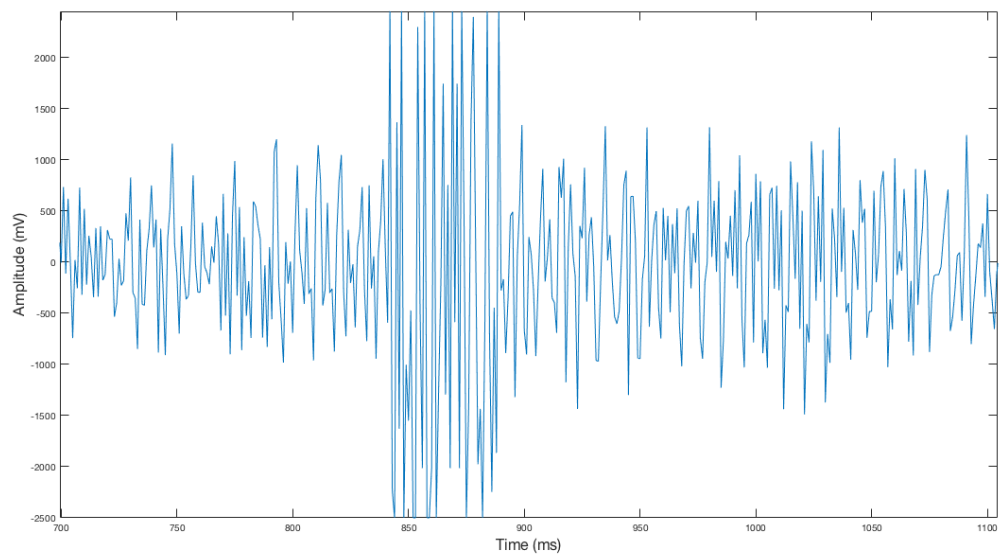


Figure 60. Shot 8 wirelessly received with our system (without ADC gain) (with pre-ADC conditioning) (0m).



Figure 61. Shot 9 from the Geophysics department system.

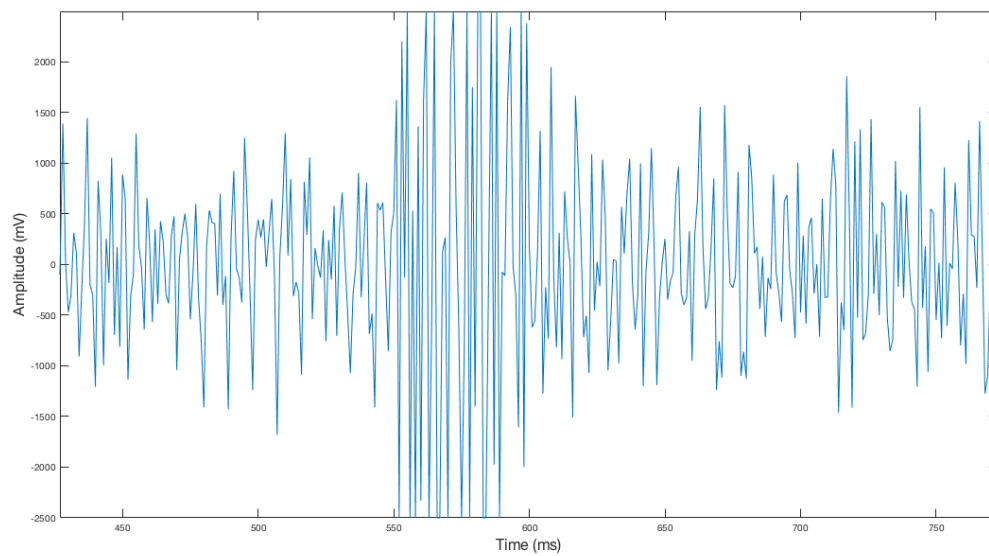


Figure 62. Shot 9 wirelessly received with our system (without ADC gain) (with pre-ADC conditioning) (50m).

6.3 Specification Verification Check List

- The achieved sampling was programmed by the ADC to exactly 2ms and elimination of any other related delays (code machine cycles and wireless communication, ...etc.) was done.
- Complete 24 bits without any missing bits.
- Wireless communication was successfully verified within a range of more than 50m
- Total price of the implemented prototype did not exceed 500 SR (around 400 SR).

Chapter 7

Conclusions and Recommendations

7.1 Conclusions

In short, a complete wireless seismic data acquisition prototype was implemented. Signals from a connected sensor enters the system differentially through conditioning circuits that magnify, filter and convert the signals into digital words. These digits are then transmitted wirelessly to another element (receiver) which can store and plot the obtained signals. A microcontroller was used to manage the ADC and wireless communication operations. Design improvement have been made to obtain better performance. For example, using a band pass filter on received signal can significantly enhance the data. Wireless transmission speed was affected by the microcontroller programming code (machine cycles delay). Also, noisy signals are received when high frequency signals are wirelessly transmitted. For the signal conditioning, the ADC programmable gain was sufficient to obtain readable signals as can be seen in Figure 46, Figure 50, Figure 52, Figure 54, Figure 56 and Figure 58 from the final the design verification detailed results (Section 6.2). It was a great opportunity to experience a complex project design and build a complete working prototype.

7.2 Recommendations

According to the prototype provided so far, it is noticed that wireless communication data rate is affected by the code period. Thus, for better and efficient communication, it is better to use simpler language that does the necessary tasks with the proper order. Assembly language is a good but complicated solution for this. Here are further recommendations below:

- Build more efficient programming code of the microcontroller to obtain faster data rate.
- Using SPI serial interface instead of UART to utilize the maximum possible data rate of Xbee S2C, which is 5Mbps
- When expanding this final design to multi transmission points, Xbee S2C is recommended for its optimized multi-points in a network

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Appendix A

MATLAB scripts of the sensors models

```
%Coil based geophone sensor response
clc
clear
close all
% Initialization
lambda = 0.25; % Damping ratio (relative to the critical damping)
w0 = 10; % Resonant frequency
Cs = 1; % Sensitivity constant
% Transfer Functions
H_displacement = tf([1 0 0 0],[1 2*lambda*w0 w0^2]);
H_velocity = tf([1 0 0],[1 2*lambda*w0 w0^2]);
H_acceleration = tf([1 0],[1 2*lambda*w0 w0^2]);
% Bode plots
figure(1)
bode(H_displacement);
title('Displacement to Electrical Signal Transfer Function');
figure(2)
bode(H_velocity);
title('Velocity to Electrical Signal Transfer Function');
figure(3)
bode(H_acceleration);
title('Acceleration to Electrical Signal Transfer Function');

%% MEMS sensor response
clc
clear
close all
% Initialization
lambda = 0.2; % Damping ratio (relative to the critical damping)
w0 = 10; % Resonant frequency
Cs = 1; % Sensitivity constant
% Transfer Functions
H_displacement = tf([1 0 0],[1 2*lambda*w0 w0^2]);
H_velocity = tf([1 0],[1 2*lambda*w0 w0^2]);
H_acceleration = tf([1],[1 2*lambda*w0 w0^2]);
% Bode plots
figure(1)
bode(H_displacement);
title('Displacement to Electrical Signal Transfer Function');
figure(2)
bode(H_velocity);
title('Velocity to Electrical Signal Transfer Function');
figure(3)
bode(H_acceleration);
title('Acceleration to Electrical Signal Transfer Function');
```

Appendix B

MATLAB scripts of the FFT signal processing

```
%%*****Apply FFT the optimized signal*****%%
%%**Find the peak frequency then store it into R_mean array**%%
[~,f_index_mean]=max(Y_mean);
R_mean(i)=abs(f_axis_mean(f_index_mean)); %
%%**Find the peak frequency then store it into R_mean array**%%
%plot FFT of the signal
figure(s);subplot(5,3,index);plot(f_axis_mean,Y_mean);xlim([rangeSta
rt(i) rangeStop(i)]);
%%*****Apply FFT to band passed signal*****%%
f_axis_BPF=-Fs/2:Fs/length(Signal_BPF{i}):Fs/2-
Fs/length(Signal_BPF{i});
Y_BPF = abs(fftshift(fft(Signal_BPF{i})));
%%*****Apply FFT to band passed signal*****%%
%%**Find the peak frequency then store it into R_BPF array**%%
[m,f_index_BPF]=max(Y_BPF);
R_BPF(i)=abs(f_axis_BPF(f_index_BPF));
%%**Find the peak frequency then store it into R_BPF array**%%
%plot FFT of the signal
figure(s+1);subplot(5,3,index);plot(f_axis_BPF,Y_BPF);xlim([rangeSta
rt(i) rangeStop(i)]);
index=index+1; %increase the index to continue plotting
end
diff=R_mean-R_BPF %to find how far do measured frequencies agree
(with and without BPF)
R_mean %display the final results
R_BPF %display the final results
```

Appendix C

Arduino code of the final prototype

```
/*
  AD7714 (Delta-Sigma ADC) testing code
  March 2018
  By. Abdullah Al-Nafisah and Mojtaba Al-Shams
  External connections:
  PIN 1 -> SCLK : external clock SPI Arduino (pin 52 MEGA) (pin 13
nano) Arduino
  PIN 2 -> MCLK IN : 2.4576 MHz by crystal (input)
  PIN 3 -> MCLK OUT : 2.4576 MHz (output)
  PIN 4 -> POL = 0: first transition of the serial clock in a data
transfer operation is from a low to a high
  PIN 5 -> (SYNC) = 1 : synchronization of the digital filters and
analog modulators
  PIN 6 -> (RESET) : Reset (PIN 49 MEGA) (pin 9 nano) Arduino
  PIN 7/8/9/10 -> AIN1/AIN2/AIN3/AIN4 : Analog Input Channel
  PIN 11 -> (STANDBY) = 1 : disable Standby
  PIN 12 -> AVdd = 5 V
  PIN 13 -> BUFFER = 0 : analog input is shorted out
  PIN 14 -> REF IN(-) = AGND : negative input of the differential
reference input
  PIN 15 -> REF IN(+) = 2.5 V : positive input of the differential
reference input
  PIN 16/17 -> AIN5/AIN6 : Analog Input Channel
  PIN 18 -> AGND = GND
  PIN 19 -> (CS) : chip select SPI (PIN 53 MEGA) (pin 10 nano)
Arduino
  PIN 20 -> (DRDY) : logic input of the AD7714 (PIN 48 MEGA) (pin 6
nano) Arduino
  PIN 21 -> DOUT : serial data output, MISO SPI (PIN 50 MEGA) (pin
12 nano) Arduino
  PIN 22 -> DIN : serial data input, MOSI SPI (PIN 51 MEGA) (pin 11
nano) Arduino
  PIN 23 -> DVdd = 5 V
  PIN 24 -> DGND = GND
*/
//-----
// Used Libraries
#include <SPI.h>
#include <SD.h>
#include <SoftwareSerial.h>
//-----
// Timer
int c=0;
int i = 0;
int Stop;
long StartTimer;
long FinishTimer;
long TotalTime;
//-----
// ADC parameters
const int ADC_CS = 10;
const int Reset = 9;
const int DRDY = 6;
unsigned long Data;
signed long Volt;
```

```

byte D[3];
SPISettings ADCsettings(500000, MSBFIRST, SPI_MODE3);
//-----
// wireless Communication parameters
int DD;
const int Rx = 3, Tx = 2;
SoftwareSerial mySerial = SoftwareSerial(Rx, Tx);
//-----
// SD Card parameters
//const int SD_CS = 4;
//-----
//-----
void setup() {
  Serial.begin(2000000);
  // wait for serial port to connect
  while (!Serial) {}
  //-----
  // Define pins
  pinMode(DRDY, INPUT);
  pinMode(Reset, OUTPUT);
  pinMode(ADC_CS, OUTPUT);
  digitalWrite(Reset, HIGH);
  digitalWrite(ADC_CS, HIGH);
  //-----
  // wireless Communication
  // Set the data rate
  mySerial.begin(19200);
  Serial.println("wireless Seismic Acquisition, Serial
Communication is ON");
  mySerial.println("Hello, Transmitter is talking");
  //-----
  // ADC
  // wake up the SPI bus
  SPI.begin();
  // Resetting 1
  digitalWrite(ADC_CS, LOW);
  digitalWrite(Reset, LOW);
  delay(100);
  digitalWrite(Reset, HIGH);
  digitalWrite(ADC_CS, HIGH);
  // Resetting 2
  SPI.beginTransaction(ADCsettings);
  digitalWrite(ADC_CS, LOW);
  SPI.transfer(0xFF);
  SPI.transfer(0xFF);
  SPI.transfer(0xFF);
  SPI.transfer(0xFF);
  digitalWrite(ADC_CS, HIGH);
  SPI.endTransaction();
  // Configuring
  SPI.beginTransaction(ADCsettings);
  digitalWrite(ADC_CS, LOW);
  SPI.transfer(0x26); // Filter (high) (Differential 26) (Single
ended 20)
  SPI.transfer(0x40); //
  SPI.transfer(0x36); // Filter (low) (Differential 36) (Single
ended 30)
  SPI.transfer(0x26); //
  SPI.transfer(0x16); // Mode (Differential 16) (Single ended 10)

```

```

SPI.transfer(0x20); // With gain 3C, without 20
digitalWrite(ADC_CS, HIGH);
SPI.endTransaction();
delay(100);
//-----

// SD Card
pinMode(SD_CS, OUTPUT);
digitalWrite(SD_CS, HIGH);
SPI.beginTransaction(SDsettings);
digitalWrite(SD_CS, LOW);
Serial.print("Initializing SD card...");
// see if the card is present and can be initialized:
if (!SD.begin(SD_CS)) {
  Serial.println("Card failed, or not present");
  // don't do anything more:
  return;
}
Serial.println("card initialized.");
digitalWrite(SD_CS, HIGH);
SPI.endTransaction();
//-----

}
//-----

//-----

void loop() {
  Stop = digitalRead(5);
  FinishTimer = millis();
  while (Stop == 1)
  {
    if (i==1)
    {
      FinishTimer = millis();
      Serial.println("-----");
      Serial.print("Start Time: ");
      Serial.println(StartTimer);
      Serial.print("Finish Time: ");
      Serial.println(FinishTimer);
      TotalTime = (FinishTimer-StartTimer);
      Serial.print("Total Time (in milli-seconds): ");
      Serial.println(TotalTime);
      Serial.print("-----");
      i = 0;
    }
    c = 0;
    Stop = digitalRead(5);
  }

  if (i==0)
  {
    StartTimer = millis();
    i = 1;
    Serial.println("-----");
  }
  //-----

  // ADC part
  if (digitalRead(DRDY) == LOW) {
    SPI.beginTransaction(ADCsettings);
    digitalWrite(ADC_CS, LOW);
    SPI.transfer(0x5E); // (Differential 5E) (Single ended 58)
    D[0] = SPI.transfer(0x5E); // (Differential 5E) (Single ended
58)

```

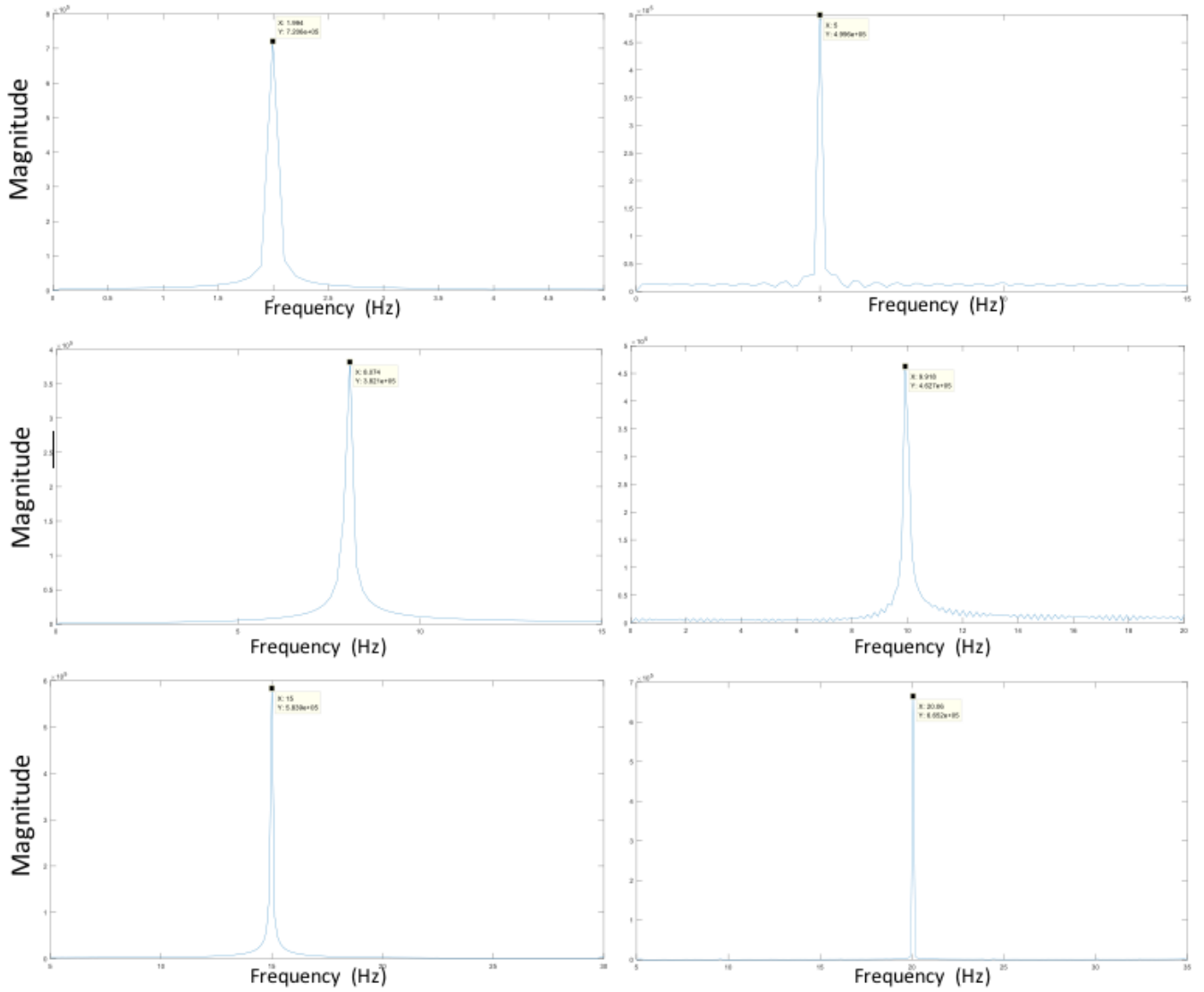
```

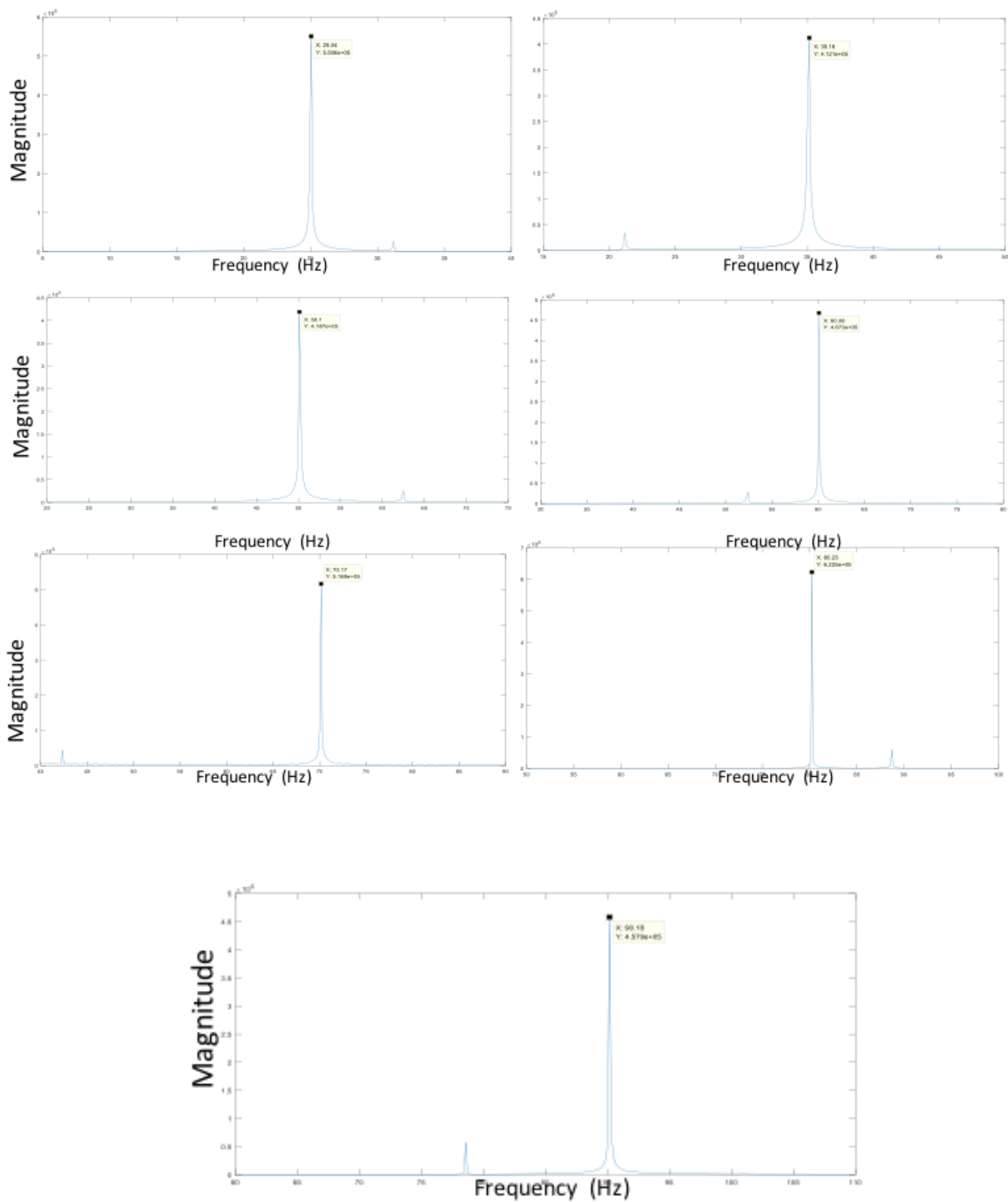
58) D[1] = SPI.transfer(0x5E); // (Differential 5E) (Single ended
58) D[2] = SPI.transfer(0x5E); // (Differential 5E) (Single ended
digitalWrite(ADC_CS, HIGH);
SPI.endTransaction();
//-----
// Calculations
Data = (float)D[2] + (float)D[1] * 256 + (float)D[0] * 256 *
256;
volt = 1000.0 * (Data / 3355443.2) - 2500.0;
Serial.println(volt);
//-----
// wireless Communication
DD = (int)(volt + 2500);
mySerial.println(DD);
//-----
// SD Card
// open the file. note that only one file can be open at a time,
// so you have to close this one before opening another.
SPI.beginTransaction(SDsettings);
digitalWrite(SD_CS, LOW);
File dataFile = SD.open("datalog.txt", FILE_WRITE);
// if the file is available, write to it:
if (dataFile) {
dataFile.println(volt);
dataFile.close();
// print to the serial port too:
}
// if the file isn't open, pop up an error:
else {
Serial.println("error opening datalog.txt");
}
digitalWrite(SD_CS, HIGH);
SPI.endTransaction();
}
//-----
}
//-----
//-----
//-----

```

Appendix D

FFT plots of the wireless received signals





Appendix E

Circuits components and corresponding calculations

Name	measured	Ideal			tolerence (%)	STAGE	CONNECTION (PIN)	COMMENTS		
R1	0.984k	1K	1.00E+03	9.84E+02	1.60	PREAMPLIFIER	Vin+ TO 2	GAIN	CALCULATED	-9.939
R2	0.984k	1K	1.00E+03	9.84E+02	1.60	PREAMPLIFIER	Vin- TO 3		MEASURED	-9.946
R3	9.78K	10K	1.00E+04	9.78E+03	2.20	PREAMPLIFIER	2 TO 1	OUTPUT AT PIN 1		
R4	9.78K	10K	1.00E+04	9.78E+03	2.20	PREAMPLIFIER	3 TO GND	FREE PINS 5 TO 7		
C1		100n				PREAMPLIFIER	8 TO GND			
C2		100n				PREAMPLIFIER	4 TO GND			
					4.00					
RLPF1	8.26K	8.2K	8.20E+03	8.26E+03	0.73	LOW-PASS-FILTER	INPUT TO X	GAIN		1.00
RLPF2	14.92K	15K	1.50E+04	1.49E+04	0.53	LOW-PASS-FILTER	X TO 3	FLPF (Hz)	CALCULATED	966.57
CLPF1		22n		2.20E-08		LOW-PASS-FILTER	X TO 6		MEASURED	1.025k
CLPF2		10n		1.00E-08		LOW-PASS-FILTER	3 TO GND	OUTPUT AT PIN 6		
C3		100n				LOW-PASS-FILTER	7 TO GND	FREE PINS 1, 5 & 8		
C4		100n				LOW-PASS-FILTER	4 TO GND			
RG Cref	9.8K	10K	1.00E+04	9.80E+03	2.00	GAIN CONTROL	2 TO GND	GAIN		
RG C1	99.6K	100K	1.00E+05	9.96E+04	0.40	GAIN CONTROL	2 TO 6	OPTION	CALCULATED	MEASURED
RG C2	683K	680K	6.80E+05	6.83E+05	0.44	GAIN CONTROL	2 TO 6	GC1	10.163	10.847
RG C3	.998M	1M	1.00E+06	9.98E+05	0.20	GAIN CONTROL	2 TO 6	GC2	69.694	70.653
RG C4	1.995M	2.2M	2.20E+06	2.00E+06	9.32	GAIN CONTROL	2 TO 6	GC3	101.837	102.895
C5		100n				GAIN CONTROL	7 TO GND	GC4	203.571	205.326
C6		100n				GAIN CONTROL	4 TO GND	OUTPUT AT PIN 6		
								FREE PINS 1, 5 & 8		
RHPF	9.77K	10K	1.00E+04	9.77E+03	2.30	HIGH-PASS FILTER		FHPF (Hz)	CALCULATED	3.47
CHPF		4.7u		4.70E-06		HIGH-PASS FILTER			MEASURED	3.876
R5	9.76K	10K	1.00E+04	9.76E+03	2.40	CLIPPER		TOTAL BW	CALCULATED	963.11
R6		100	100	99.7	0.30	CLIPPER			MEASURED	1021.12
R7		10	10	10.2	2.00	CLIPPER		TOTAL GAIN		
C7		10n				CLIPPER		MINIMUM		-101.0129
								MAXIMUM		-2023.301
	Possible options for Gain control stage							MAXIMUM INPUT		
	#	R1	R2	R3	R4	Req	Gain	IDEAL MAXIMUM		24.749 mV
	1	0	0	0	0	-	-	IDEAL MINIMUM		1.236 mV
	2	1	0	0	0	1.00E+05	11.000			
	3	0	1	0	0	6.80E+05	69.000			
	4	1	1	0	0	8.72E+04	9.718			
	5	0	0	1	0	1.00E+06	101.000			
	6	1	0	1	0	9.09E+04	10.091			
	7	0	1	1	0	4.05E+05	41.476			
	8	1	1	1	0	8.02E+04	9.019			
	9	0	0	0	1	2.20E+06	221.000			
	10	1	0	0	1	9.57E+04	10.565			
	11	0	1	0	1	5.19E+05	52.944			
	12	1	1	0	1	8.39E+04	9.386			
	13	0	0	1	1	6.88E+05	69.750			
	14	1	0	1	1	8.73E+04	9.730			
	15	0	1	1	1	3.42E+05	35.186			
	16	1	1	1	1	7.74E+04	8.737			

Appendix F

MATLAB scripts of the signal conditioning circuit without ADC

```
clear all
close all

Preamplifier_gain=10;

R1_LPF=8.2e3;
R2_LPF=15e3;

C1_LPF=22e-9;
C2_LPF=10e-9;

Gain_selected= 11; %[The main 4 options:11, 69, 101 and 221]

R_HPF=10e3;
C_HPF=4.7e-6;

s=tf('s');
wo_LPF=1/[sqrt(R1_LPF*R2_LPF*C1_LPF*C2_LPF)];
fo_LPF=wo_LPF/(2*pi) % the cut-off frequency of the LPF
Q_LPF=1/[wo_LPF*(R1_LPF*C1_LPF+R2_LPF*C1_LPF)] % the cut-off
frequency of the LPF
H_LPF=1/[s^2+wo_LPF/Q_LPF*s+wo_LPF^2] % the frequency response of
the LPF

wo_HPF=1/(R_HPF*C_HPF);
fo_HPF=wo_HPF/(2*pi) % the cut-off frequency of the HPF
H_HPF=s/[s+wo_HPF] % the frequency response of the HPF

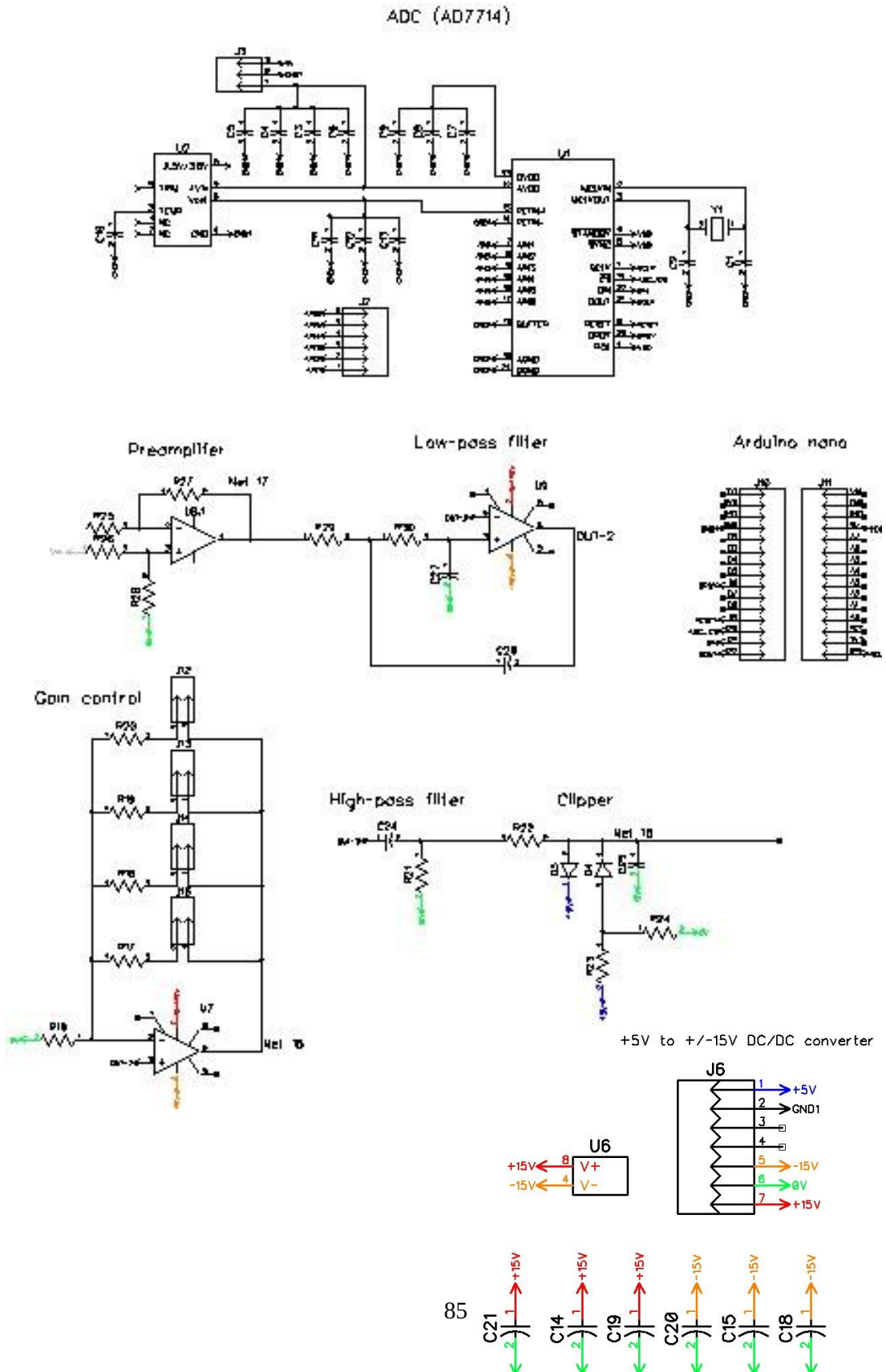
BW=fo_LPF-fo_HPF % the bandwidth of the S.C.C wo ADC
H=Preamplifier_gain*H_LPF*Gain_selected*H_HPF % the frequency
response of the S.C.C wo ADC
%%
figure(1)
bode(H_LPF) % plot of the ideal low pass filter
title('Bode plot of Sallen-Key second order low filter')
legend('cutoff frequency of 967.5117Hz')

figure(2)
bode(H_HPF) % plot of the ideal high pass filter
title('Bode plot of first order passive high filter')
legend('cutoff frequency of 3.3863Hz')

figure(3)
bode(H) % plot of the ideal signal conditioning circuit without ADC
title('Bode plot of the signal conditioning circuit without the
ADC')
legend('Bandwidth of 964.1254Hz')
```

Appendix G

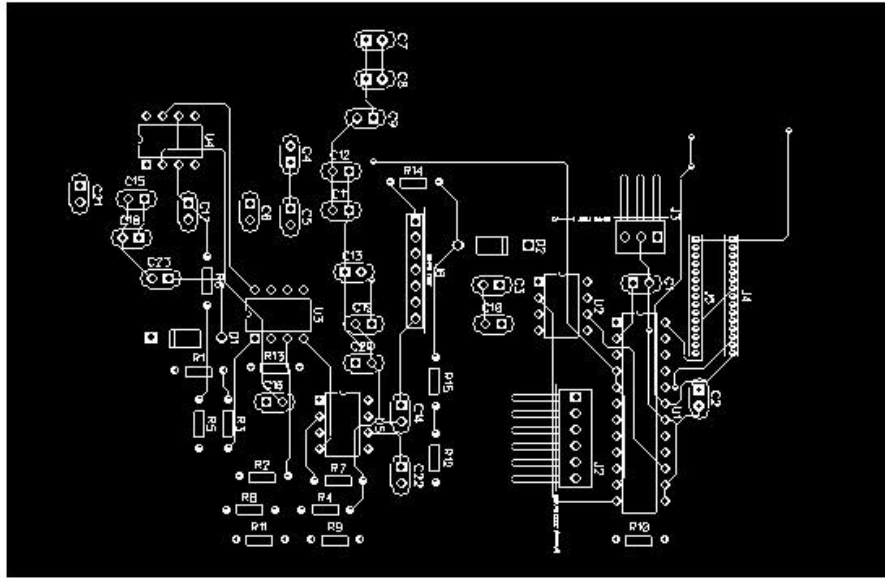
System schematic of the PCB



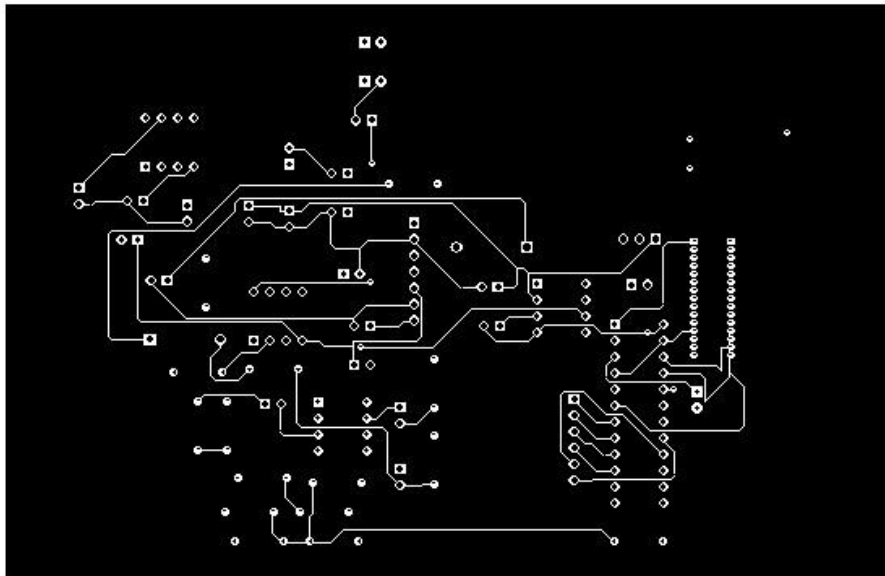
Appendix H

PCB Layout

Top



Bottom



Appendix I

Op-amps comparison

Parameters	OP07(CP)	NE532(P)	OPA2705(PA)	OPA277(PA)	OPA27(CP)	AD620(ANZ)
Power dissipation (mW)	80-150	-	-	-	500	650
Voltage Supply(V)	$\pm 3 - \pm 18$	$\pm 5 - \pm 15$	$\pm 2 - \pm 6$	$\pm 2 - \pm 18$	$\pm 4 - \pm 22$	$\pm 2.3 - \pm 18$
CMRR (dB)	100-120	70-100	66-77	115-140	100-122	73-130
Input offset voltage (μV)	60-80	500-5000	$\pm 500 - \pm 5000$	$\pm 20 - \pm 50$	$\pm 25 - \pm 100$	30-125
Input offset current (nA)	0.8-1.6	10-200	$\pm 0.0005 - \pm 0.01$	$\pm 0.5 - \pm 4$	10-75	0.3-10
Input bias current (nA)	$\pm 1.8 - \pm 2.2$	200-1000	$\pm 0.001 - \pm 0.01$	$\pm 0.5 - \pm 4$	$\pm 15 - \pm 80$	0.5-2
Unity-gain bandwidth (MHz)	0.4-0.6	10	1	1	5-8	1
Input resistance [differential ($M\Omega \parallel pF$) common ($G\Omega \parallel pF$)]	8-33 $\parallel$ - --	0.3 $\parallel$ - --	4000 $\parallel$ 4 5000 $\parallel$ 4	100 $\parallel$ 3 250 $\parallel$ 3	-- 2 $\parallel$ 2.5	10000 $\parallel$ 2 10 $\parallel$ 2
Output impedance (Ω)	-	0.3	-	40	70	-
Input offset voltage ($nV / \sqrt{Hz}$) [10-1kHz]	10.5-9.8	8-5	45	12-8	8-3.2	9-13
Equivalent input noise current ($nA / \sqrt{Hz}$) [10-1kHz]	0.35- 0.13	0.0027-0.0007	$2.5 * 10^{-6}$	0.0002	0.0017-0.0004	$100 * 10^{-6}$
Slew rate (V/ μs)	0.3	9	0.6	0.8	1.7-1.9	0.75-1.2
Cost (\$)	0.894	1.025	2.426	3.416	3.553	10.213